Querying Cohesive Subgraphs in Temporal Graphs: A Unified and Efficient Framework

Yinyu Liu
Harbin Institute of Technology, Shenzhen
China
23S151053@stu.hit.edu.cn

Kaiqiang Yu
Nanyang Technological University
Singapore
kaiqiang002@e.ntu.edu.sg

Shengxin Liu
Harbin Institute of Technology, Shenzhen
China
sxliu@hit.edu.cn

Cheng Long
Nanyang Technological University
Singapore
c.long@ntu.edu.sg

Xun Zhou
Harbin Institute of Technology, Shenzhen
China
zhouxun2023@hit.edu.cn

ABSTRACT

Temporal graphs are widely used to model dynamic interactions in various domains, including social networks, biology, and e-commerce. Querying *cohesive subgraph models (CSMs)* within a given time interval in temporal graphs is a fundamental problem in graph analytics. While numerous index-based algorithms have been proposed, they all focus on querying one specific CSM. As a result, these index techniques for querying one type of CSM cannot be adopted for querying the others. In this paper, we focus on querying a generalized class of *non-overlapping* and *monotone* CSMs, including historical connected component (His-CC) and historical k-core (His-Core), and study the corresponding query problem called *His-CSM query*. To solve the problem, we propose a *unified* index-based framework. Specifically, we transform His-CSM queries with respect to various CSMs into a unified query problem, namely *spanning connected component query*, and introduce an index-based solution to efficiently answer this newly formulated query. Finally, we adopt our framework to answer both His-CC and His-Core queries. Extensive experiments are conducted on real-world graphs, and the results show that our method outperforms the state-of-the-arts by up to 60× faster for His-CC query and 8,000× faster for His-Core query.

PVLDB Reference Format:
Yinyu Liu, Kaiqiang Yu, Shengxin Liu, Cheng Long, and Xun Zhou. Querying Cohesive Subgraphs in Temporal Graphs: A Unified and Efficient Framework. PVLDB, 14(1): XXX-XXX, 2020.
doi:XX.XX/XXX.XX

PVLDB Artifact Availability:
The source code, data, and/or other artifacts have been made available at https://anonymous.4open.science/r/Querying-Cohesive-Subgraphs-in-Temporal-Graphs-C38D.

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Proceedings of the VLDB Endowment, Vol. 14, No. 1 ISSN 2150-8097.
doi:XX.XX/XXX.XX

1 INTRODUCTION

Temporal graphs are widely used to capture evolving interactions between entities, with each edge being associated with a timestamp that indicates the interaction time [5, 33, 35, 48]. Examples include social interactions, financial transactions, biological systems, and more. Querying historical cohesive subgraphs, which aims to extract cohesive subgraphs from a graph snapshot defined by a given query time interval, is a fundamental problem in temporal graph analytics and has been extensively studied in the literature [10, 24, 26, 32, 40–42, 49]. Specifically, cohesive subgraphs, which represent dense regions in a graph, often carry significant information and are of substantial interest in real-world applications. In recent years, numerous cohesive subgraph models (CSMs) have been proposed [8, 16, 22, 59], including models specifically designed for temporal graphs [21, 28, 37, 54, 60]. Querying historical cohesive subgraphs across various graph snapshots enables tracking the evolution of dense regions, thereby revealing long-term network behaviors and uncovering significant patterns or anomalies.

The problem of querying historical cohesive subgraphs has broad applications across various disciplines, including financial anomaly detection [4, 36, 58], temporal-aware recommendation systems [53], infectious disease tracking [13], and so on. As an example of financial anomaly detection, sudden changes in tightly integrated trading groups usually correspond to fraudulent activities, market manipulation, or other forms of financial misconduct. Thus, tracking cohesive subgraphs over time in financial networks can help identify these unusual patterns [4, 36]. As a second example of infectious disease tracking, temporal graphs can model the spread of infectious diseases by representing contact networks between individuals [13]. Cohesive subgraphs in these graphs might correspond to clusters of individuals who are more likely to spread the disease within a certain period, providing key insights for outbreak control. Furthermore, the authors in [54, 60] conduct case studies by querying historical cohesive subgraphs, such as historical k-cores and historical connected components, to detect changes in research groups using graphs constructed from DBLP data.

Existing methods and limitations. One naive online solution for historical CSMs queries is to first collect all the edges whose timestamps fall within the query time interval and then run a corresponding CSM mining algorithm on the resulting graph. However,

this approach is highly inefficient especially when dealing with large query time intervals or large graphs, as it requires accessing and processing all relevant edges, which can be computationally expensive. Therefore, existing studies solely targets on querying one specific CSM and focus on designing index-based solutions that leverage the unique properties of the targeted CSM. Specifically, index structures are utilized to store useful information that supports efficient historical queries for one specific CSM, e.g., historical connected component (His-CC) [54] and historical k-core (His-Core) queries [46, 60]. Nevertheless, these indexing approaches are restricted to certain cohesive subgraph models. In other words, techniques designed for querying one type of historical CSM cannot be directly applied to the others. As a result, this lack of generality introduces significant additional overhead. Specifically, for each different historical CSM, index-based solutions need to be designed from scratch; and graph database systems need to consider a variety of index schemes to support diverse historical CSM queries.

Our unified framework. To address the aforementioned limitations, it is necessary to answer the following research question – "*Can we design a unified index-based framework to support historical CSM queries for various CSMs?*" Motivated by this, in this paper, we focus on a generalized class of *non-overlapping* and *monotone* CSMs and study a general query problem, referred to as the *His-CSM query*, which involves querying historical non-overlapping and monotone CSMs. Here, the non-overlapping and monotone properties are commonly shared by many CSMs, including connected component, k-core, and k-edge-connected component. Formally, given a temporal graph $\mathcal{G}$, a CSM $\mathcal{P}$, and a query time interval $[t_s, t_e]$, the His-CSM query seeks to identify all subgraphs within the graph snapshot corresponding to $[t_s, t_e]$ that satisfy $\mathcal{P}$. In contrast to existing studies [46, 54, 60], this newly formulated problem is generalized to accommodate a wide range of CSMs.

To solve the problem, we propose two unified index-based frameworks, namely `AF-3DT` and `AF-BTAS`. Among them, the best one `AF-BTAS` achieves better theoretical and practical performance compared to the state-of-the-art methods `TSF` and `PHC` [54, 60], when applied to querying His-CC and His-Core (as summarized in Table 1 and verified through our experiments in Section 4). In particular, for His-CC queries, our `AF-BTAS-CC` builds the index in $O(m \log m)$ and answers the query in $O(n + \log^2 m)$ time, which costs less time than the SOTA `TSF` [54], i.e., $O(mt_{\max})$ and $O(n + t_s)$. Here, m and n are the number of edges and vertices in the temporal graph, respectively; $t_{\max}$ is the maximum timestamp with $t_{\max} \leq m$ and t_s is the starting timestamp of the query time interval $[t_s, t_e]$, which is bounded by $t_{\max}$. In summary, our `AF-BTAS` enjoys both generality and efficiency – it generalizes to support various CSMs while outperforming the SOTAs for specific CSMs. Specifically, our framework involves the following two components.

The first component is a *transformer*, which transforms the His-CSM query for various CSMs in the temporal graph into a unified query formulation, namely the *spanning connected component (Spanning-CC) query* in the *affiliated (spanning) graph*. Answering the Spanning-CC query is equivalent to solving the original His-CSM query, thereby ensuring the generality of our framework in addressing different His-CSM queries. The second component corresponds to two index-based solutions, namely `AF-3DT` and `AF-BTAS`, which are based on a novel concept of *affiliated forests (AF)*. These index-based solutions are specifically designed to efficiently answer the newly formulated Spanning-CC query.

Our contributions. Our main contributions are as follows:

- We formulate a novel query problem of querying historical non-overlapping and monotone cohesive subgraphs. (Section 2)
- We propose two unified index-based solutions called `AF-3DT` and `AF-BTAS` for answering His-CSM queries. Further, we apply our `AF-BTAS` to answering His-CC and His-Core queries. (Section 3)
- We conduct extensive performance evaluations using real-world datasets, and the experimental results demonstrate the superiority of our proposed method. (Section 4)

In addition, we discuss related work in Section 5 and conclude the paper in Section 6.

2 PRELIMINARIES

We focus on an undirected temporal graph $\mathcal{G} = (V, \mathcal{E})$, where V denotes the set of n vertices and $\mathcal{E}$ denotes the set of m *temporal* edges. Here, each temporal edge is represented by a triple (u, v, t), where $u \in V$, $v \in V$, and $t \in \mathbb{N}$ indicates the timestamp at which u interacts with v. Following existing studies [21, 28, 37, 54, 60], we assume that all timestamps are consecutive integers ranging from 1 to $t_{\max}$, thereby $t_{\max} \leq |\mathcal{E}|$. Given a time window $[t_s, t_e]$, the *projected graph* of $\mathcal{G}$ over $[t_s, t_e]$, denoted by $G_{[t_s,t_e]} = (V, E_{[t_s,t_e]})$, is an undirected graph that includes the set of vertices V and the set of edges $E_{[t_s,t_e]} = \{(u, v) \mid (u, v, t) \in \mathcal{E}, t \in [t_s, t_e]\}$. We remark that projected graphs ignore the temporal information associated with the edges. Given a graph $G = (V, E)$ and a vertex set $V_{sub} \subseteq V$, we use $G[V_{sub}]$ to denote the subgraph induced by V_{sub}, i.e., $G[V_{sub}]$ includes the set of vertices V_{sub} and the set of edges $\{(u, v) \in E \mid u, v \in V_{sub}\}$. We remark that all the subgraphs considered in this paper are by default vertex-induced subgraphs.

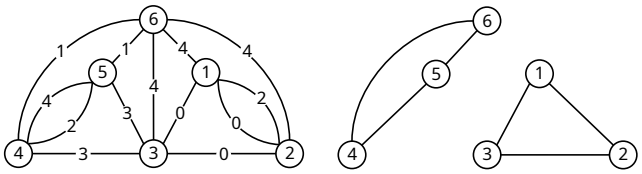
Let $\mathcal{P}$ be a *cohesive (vertex-induced) subgraph model* (CSM) defined on an undirected graph, e.g., connected component, k-core, clique, k-plex, etc. Given an undirected graph $G = (V, E)$, we define $\mathcal{P}(G)$ as a set of (vertex-induced) subgraphs of G. For a subgraph g of G, if $g \in \mathcal{P}(G)$ (resp. $g \notin \mathcal{P}(G)$), we say that g satisfies (resp. violates) CSM $\mathcal{P}(G)$. In this paper, we focus on those CSMs that are both *non-overlapping* and *monotone*.

Definition 2.1 (Non-overlapping). A CSM $\mathcal{P}$ is said to be *non-overlapping* iff for a graph $G = (V, E)$ and any two subgraphs $G[V_1] \in \mathcal{P}(G), G[V_2] \in \mathcal{P}(G)$ with $V_1 \subseteq V, V_2 \subseteq V$, the vertex sets V_1 and V_2 are disjoint, i.e., $V_1 \cap V_2 = \emptyset$.

Clearly, for a non-overlapping CSM $\mathcal{P}$, $\mathcal{P}(G)$ consists of at most $|V|$ different subgraphs. For example, the connected component belongs to a non-overlapping CSM and thus there are at most $|V|$ different connected components in a graph G.

Definition 2.2 (Monotone). A CSM $\mathcal{P}$ is said to be *monotone* iff given a graph $G = (V, E)$, for any subgraph $G[V_{sub}] \in \mathcal{P}(G)$ (with $V_{sub} \subseteq V$) and any supergraph $G' = (V', E')$ of G (with $V \subseteq V'$ and $E \subseteq E'$), there exists a subgraph $G'[V'_{sub}]$ in $\mathcal{P}(G')$ such that $V_{sub} \subseteq V'_{sub}$.

Intuitively, for a monotone CSM $\mathcal{P}$, all vertices in a subgraph satisfying $\mathcal{P}(G)$ remain in the same subgraph satisfying $\mathcal{P}(G)$ after



(a) A temporal graph $\mathcal{G}$ (b) The projected graph $G_{[0,2]}$

Figure 1: Illustration the basic concepts of temporal graph and projected graph

adding additional vertices and/or edges to G. Besides, we note that quite a few CSMs satisfy the non-overlapping and monotone properties, including connected component and k-core. Formally, we have the following lemma.

Lemma 2.3. *The connected component, k-core, connected k-core are both non-overlapping and monotone.*

Proof. The proof is in Appendix of the technical report [1]. □

To ease the writing, we refer to the CSMs considered in this paper as those that satisfy both the non-overlapping and monotone properties by default. We note that the mentioned CSMs are originally defined on undirected and unweighted graphs. To extend them to temporal graphs, there are several studies in the literature, leading to various temporal versions of the same non-temporal CSMs. Following existing studies [54, 60], we focus on one such extension, namely, *historical-CSMs (His-CSMs)*. Intuitively, a His-CSM is defined based on a CSM $\mathcal{P}$ and a specified time window $[t_s, t_e]$. Formally, we have the following definition.

Definition 2.4 (Historical CSMs). Given a temporal graph $\mathcal{G}$, a time window $[t_s, t_e]$ and a CSM $\mathcal{P}$, a subgraph $\mathcal{G}[V_{sub}]$ of $\mathcal{G}$ satisfies the $\mathcal{P}$-based His-CSM over $[t_s, t_e]$ if and only if the projected graph of $\mathcal{G}[V_{sub}]$ over $[t_s, t_e]$ satisfies $\mathcal{P}$.

Now, we are ready to formulate the problem studied in this paper.

Problem Definition (Historical Non-overlapping and Monotone Cohesive Subgraph Query). *Given a temporal graph $\mathcal{G}$, a time window $[t_s, t_e]$ and a non-overlapping and monotone CSM $\mathcal{P}$, the Historical-CSM query seeks to find all sets of vertices each of which S includes at least two vertices (i.e., $|S| \geq 2$) and induces a subgraph $G_{[t_s,t_e]}[S]$ in $\mathcal{P}(G_{[t_s,t_e]})$.*

To illustrate, we take the historical connected 2-core as an example of non-overlapping and monotone CSMs in Figure 1. Given a temporal graph in Figure 1(a), its historical connected 2-cores across different time intervals are shown in Figure 2. Specifically, each grid corresponds to a time interval and each historical connected 2-core is colored with the same color, e.g., there are two connected 2-cores in the projected graph $G_{[0,2]}$, i.e., $G[\{1, 2, 3\}]$ and $G[\{4, 5, 6\}]$, marked red and blue respectively.

We remark that the Historical-CSM query has been widely studied recently. However, all focus on one specific type of CSM, i.e., Historical-Connected-Component (or, His-CC) [54] and Historical-k-Core (or, His-Core) [60]. Thus, previous proposed methods cannot be generalized to query other CSMs. In the following, we will propose a *unified* and efficient framework to solve the Historical-CSM query and then apply the proposed method to answer His-Core and His-CC queries.

3 OUR UNIFIED INDEX-BASED FRAMEWORK

We introduce our *unified* index-based framework, which constructs novel indices to support the His-CSM query. In general, it consists of two key components. The first is a *transformer*, which transforms the His-CSM query w.r.t. various CSMs in the temporal graph to a unified query formulation, namely *spanning connected component (Spanning-CC) query* in the *affiliated (spanning) graph* (Sections 3.1). Answering the later query solves the former problem equivalently. This guarantees the generality of our framework to answer different His-CSM queries. The second corresponds to two index-based solutions, namely `AF-3DT` and `AF-BTAS`, which are based on a novel concept of *affiliated forests (AF)* and are designed to answer the newly formulated Spanning-CC query (Section 3.2 and Section 3.3). Finally, we adopt our framework to solve two types of His-CSM queries, namely His-CC query and His-Core query (Section 3.4).

3.1 A Unified Query Formulation

Before presenting our unified solution, it is necessary to first show how different His-CSM queries (w.r.t. various CSMs) can be transformed into a *unified* query formulation (note that solving the unified query formulation would answer various His-CSM queries). In general, the transformation relies on the *non-overlapping and monotone* properties. Below, we elaborate on the details.

Minimal u-v-TW. We start with the following definition.

Definition 3.1 (u-v-TW). Given a temporal graph $\mathcal{G} = (V, \mathcal{E})$ and a CSM $\mathcal{P}$. Let u and v be two vertices in V, and $[t_1, t_2]$ be a time window. $[t_1, t_2]$ is a u-v time window (or simply, u-v-TW) w.r.t. $\mathcal{P}$ iff u and v belong to the same subgraph in $\mathcal{P}(G_{[t_1,t_2]})$.

Besides, a u-v-TW $[t_1, t_2]$ is said to be *minimal* iff for any sub-interval $[t'_1, t'_2] \subset [t_1, t_2]$, u and v belong to the different subgraphs in $\mathcal{P}(G_{[t'_1,t'_2]})$. Clearly, there could be multiple minimal u-v-TW between $[0, t_{\max}]$. To illustrate, consider an example in Figure 2 based on a temporal graph in Figure 1(a). The time interval $[1, 2]$ qualifies as a minimal 5-6-TW according to the definition, as these two vertices belong to the same connected 2-core in $[1, 2]$ but not within any of its sub-intervals. Similarly, the minimal 3-4-TWs are $[2, 3]$ and $[3, 4]$. Then, we observe the following lemma.

Lemma 3.2. *Let $[t_1, t_2]$ be a (minimal) u-v-TW w.r.t. $\mathcal{P}$. Any time window $[t'_1, t'_2]$ with $[t_1, t_2] \subseteq [t'_1, t'_2]$ is a u-v-TW w.r.t. $\mathcal{P}$.*

Proof. This can be verified by the monotone property of $\mathcal{P}$ (Definition 2.2). Assume that u and v belong to a subgraph g in $\mathcal{P}(G_{[t_1,t_2]})$. We note that there exists a subgraph g' of $\mathcal{P}(G_{[t'_1,t'_2]})$ such that g is contained in g' (i.e., $V(g) \subseteq V(g')$ and $E(g) \subseteq E(g')$) since $G_{[t'_1,t'_2]}$ is a supergraph of $G_{[t_1,t_2]}$ and $\mathcal{P}$ satisfies the monotone property. Therefore, u and v belong to subgraph g' in $\mathcal{P}(G_{[t'_1,t'_2]})$ and $[t'_1, t'_2]$ is a u-v-TW. □

Affiliated spanning graph. Based on the minimal u-v-TWs, we define the *affiliated spanning graph*, denoted by $\overline{\mathcal{G}}$, of $\mathcal{G}$ as below.

Definition 3.3 (Affiliated spanning graph). Given a temporal graph $\mathcal{G} = (V, \mathcal{E})$ and a CSM $\mathcal{P}$, the *affiliated spanning graph (or affiliated*

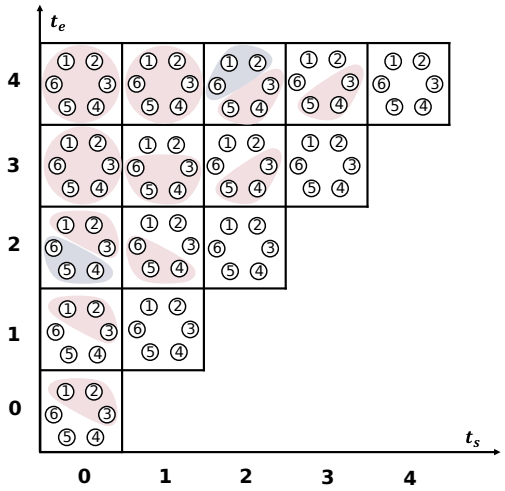


Figure 2: Illustrating historical connected 2-cores

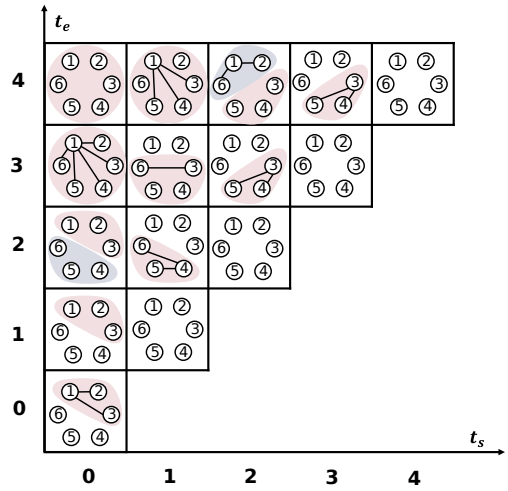


Figure 3: Illustrating the affiliated graph

graph) of $\mathcal{G}$ w.r.t. $\mathcal{P}$ is a "weighted" graph $\overline{\mathcal{G}} = (V, \overline{\mathcal{E}})$, which includes the set of vertices V and the set of *spanning edges* each of which corresponds to a minimal u-v-TW, formally,

$$\overline{\mathcal{E}} = \{(u, v, t_1, t_2) \mid u, v \in V, [t_1, t_2] \text{ is a minimal } u\text{-}v\text{-TW}\}, \quad (1)$$

and satisfies the following constraint.

> **Constraint:** For any time window $[t_s, t_e]$, two vertices u and v belong to the same subgraph in $\mathcal{P}(G_{[t_s,t_e]})$ if and only if u is reachable to v in $\overline{G}_{[t_s,t_e]}$.

Here, $\overline{G}_{[t_s,t_e]} = (V, \overline{E}_{[t_s,t_e]})$ is the *projected graph* of $\overline{\mathcal{G}}$ over $[t_s, t_e]$, which includes the set of vertices V and the set of edges, i.e.,

$$\overline{E}_{[t_s,t_e]} = \{(u, v) \mid (u, v, t_1, t_2) \in \overline{\mathcal{E}}, [t_1, t_2] \subseteq [t_s, t_e]\}. \quad (2)$$

We note that an affiliated spanning graph $\overline{\mathcal{G}}$ of $\mathcal{G}$ can be built easily by adding a spanning edge for each minimal u-v-TW in $\mathcal{G}$. We can verify this in two directions. First, if u and v belong to the same subgraph in $\mathcal{P}(G_{[t_s,t_e]})$, $[t_s, t_e]$ is a u-v-TW by definition and thus there exists at least one minimal u-v-TW $[t_1, t_2]$ such that $[t_1, t_2] \subseteq [t_s, t_e]$. In this case, $\overline{\mathcal{G}}$ has a spanning edge (u, v, t_1, t_2), and thus u and v are adjacent in $\overline{G}_{[t_s,t_e]}$. Second, u and v are reachable in $\overline{G}_{[t_s,t_e]}$ via a path $\langle u_0 = u, u_1, u_2, ..., u_t = v\rangle$ with $t \geq 1$. For a pair of vertices u_i and u_{i+1} with $0 \leq i \leq t-1$, we note that $\overline{\mathcal{G}}$ has a spanning edge (u_i, u_{i+1}, t_1, t_2) such that $[t_1, t_2] \subseteq [t_s, t_e]$, which indicates a minimal u-v-TW $[t_1, t_2]$. By Lemma 3.2, $[t_s, t_e]$ is a u-v-TW, and thus u_i and u_{i+1} belong to the same subgraph in $\mathcal{P}(G_{[t_s,t_e]})$ by definition, i.e., the *monotone property*. Finally, vertices in $\{u_0, u_1, ..., u_t\}$ (including u and v) belong to the same subgraph in $\mathcal{P}(G_{[t_s,t_e]})$ due to the *non-overlapping property*.

However, the resulting affiliated graph could be very dense. As a result, queries over a dense affiliated graph may suffer from efficiency issues. To address, an affiliated graph $\overline{\mathcal{G}}$ can be efficiently refined by removing some redundant spanning edges (u, v, t_1, t_2). We will discuss more details about the construction and refinement of affiliated graphs in the following sections.

To illustrate, see an affiliated graph in Figure 3, which is constructed based on a temporal graph in Figure 1(a). Each grid in the figure corresponds to a time interval, and an edge within a grid represents the spanning edge associated with that interval. For example, the edges $(4, 5)$ and $(4, 6)$ appear in the grid with $t_s = 1$ and $t_e = 2$, indicating that spanning edges $(4, 5, 1, 2)$ and $(4, 6, 1, 2)$ are included in the affiliated graph. There are various methods to construct such an affiliated graph, and the construction process for some specific CSM is detailed in Section 3.4.

A unified query formulation: Spanning-CC query. We observe that all vertices belonging to the same subgraph in $\mathcal{P}(G_{[t_s,t_e]})$ will induce a connected component (CC) of $\overline{G}_{[t_s,t_e]}$. Here, a CC of a graph G is a connected (vertex-induced) subgraph that is not contained in any larger connected subgraph. Further, those vertices that do not belong to any subgraph in $\mathcal{P}(G_{[t_s,t_e]})$ have no neighbors in $\overline{G}_{[t_s,t_e]}$, and thus each of these vertices corresponds to a CC of size 1. We summarize the above observation as follows, which can be easily verified based on the definition of affiliated graph.

Lemma 3.4. *For a vertex subset S of V with $|S| \geq 2$, S induces a subgraph $G_{[t_s,t_e]}[S]$ in $\mathcal{P}(G_{[t_s,t_e]})$ if and only if S induces a CC $\overline{G}_{[t_s,t_e]}[S]$ of $\overline{G}_{[t_s,t_e]}$.*

Based on the above lemma, we can answer a Historical-CSM query by solving a unified query formulation called *Spanning-CC* query, formally defined as follows.

> **Spanning-CC query:** Given an affiliated graph $\overline{\mathcal{G}}$ and a time window $[t_s, t_e]$, the spanning-CC query over $\overline{\mathcal{G}}$ aims to find all vertex sets, called spanning-CCs, where each vertex set S includes at least two vertices and induces a CC of $\overline{G}_{[t_s,t_e]}$.

We remark that the formulated spanning-CC query is *CSM-free*, as solving it does not rely on any specific input CSM. However, the construction of affiliated graphs is still CSM-specific which relies on the properties of the input CSM. Nonetheless, this step can be regarded as a one-time pre-processing for all subsequent queries.

Discussion. We note that the Historical-CC query can be regarded as a special case of the Spanning-CC query (since the former performs on a graph with temporal edges while the later performs

on a graph with spanning edges). As a result, existing methods for answering Historical-CC queries cannot be adopted for solving the newly formulated Spanning-CC query. We will present an index-based method for the later in the following sections.

3.2 Our Index-based Solution: AF-3DT

Consider an affiliated graph $\overline{\mathcal{G}}$ and a time window $[t_s, t_e]$. To answer the spanning-CC query, a simple *online* method is to find all CCs in the projected graph $\overline{G}_{[t_s,t_e]}$ using the Depth-First-Search (DFS) and the Union-Find data structure. This online method runs in $O(n+\overline{m})$, where $\overline{m} = |\overline{\mathcal{E}}|$, due to the DFS procedure. To improve efficiency, we propose an index-based solution, called AF-3DT (Affiliated-Forest-based 3-D Range Tree Index), which utilizes the novel concept of affiliated forests introduced in Section 3.2.1 and the index structure of classic 3-D range tree. Moreover, AF-3DT answers the query in $O(n + \log^2 \overline{m})$. Below, we elaborate on the details.

We observe that all CCs in a projected graph form its spanning forest, where each CC corresponds to a spanning tree. Retrieving all CCs from a spanning forest with n vertices can be done optimally in $O(n)$ via DFS (note that a spanning forest consists of at most $n-1$ edges). Hence, our idea is to *build indices for some spanning edges* in the affiliated graph so as to *efficiently identify the spanning forest of a projected graph $\overline{\mathcal{G}}_{[t_s,t_e]}$ and retrieve all CCs* when answering the spanning-CC query with a time window $[t_s, t_e]$.

3.2.1 Affiliated forests. We start with the following definition.

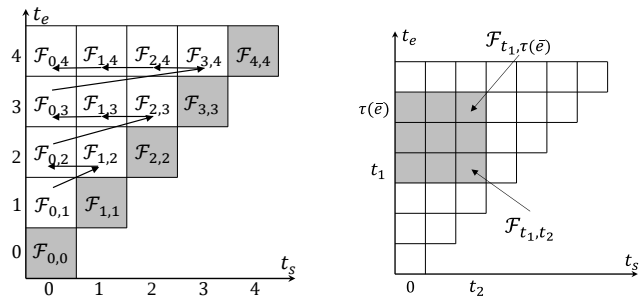
Definition 3.5 (Affiliated forest). Given an affiliated graph $\overline{\mathcal{G}}$ and a time window $[t_s, t_e]$, an affiliated forest $(V, \mathcal{F}_{t_s,t_e})$ over $[t_s, t_e]$ includes the edge set $\mathcal{F}_{t_s,t_e} \subseteq \{(u, v, t_1, t_2) \in \overline{\mathcal{E}} \mid [t_1, t_2] \subseteq [t_s, t_e]\}$ such that (1) the projected graph of $(V, \mathcal{F}_{t_s,t_e})$ over $[t_s, t_e]$ is a spanning forest of $\overline{G}_{[t_s,t_e]}$, and (2) there exists at most one spanning edge between any two vertices in $(V, \mathcal{F}_{t_s,t_e})$.

Clearly, an affiliated forest consists of at most $n-1$ edges. Thus, retrieving edges in $\mathcal{F}_{t_s,t_e}$ will help to quickly find the spanning forest of $\overline{G}_{[t_s,t_e]}$. We then show how to construct an affiliated forest for any possible time window, followed by building indices on them for efficient retrieval. Our method BuildAF has two phases, as visualized in Figure 4(a). We give the details of BuildAF below.

- **Phase-I: Initialization.** For $0 \le t \le t_{\max}$, let $(V, E_{t,t})$ be an arbitrary spanning forest of $\overline{G}_{[t,t]}$. We add to $\mathcal{F}_{t,t}$ a spanning edge (u, v, t, t) for each edge (u, v) in $E_{t,t}$, formally,

$$\mathcal{F}_{t,t} = \{(u, v, t, t) \mid (u, v) \in E_{t,t}\}. \tag{3}$$

- **Phase-II: Iterative computations.** For $0 \le t_s < t_e \le t_{\max}$, we iteratively compute $\mathcal{F}_{t_s,t_e}$ based on a certain ordering. Specifically, we have $t_{\max}$ rounds with t_e varying from 1 to $t_{\max}$, and in the i-th round (i.e., $t_e = i$), we iteratively compute $\mathcal{F}_{t_s,i}$ for t_s from $i-1$ to 0. In particular, we compute $\mathcal{F}_{t_s,t_e}$ based on the edges $(u, v, t_s, t_e) \in \overline{\mathcal{E}}$ and two sets, namely $\mathcal{F}_{t_s+1,t_e}$ and $\mathcal{F}_{t_s,t_e-1}$, previously obtained in the ordering or Phase-I, as below.
  - **Step 1.** Initialize $\mathcal{F}_{t_s,t_e}$ by $\mathcal{F}_{t_s+1,t_e}$.
  - **Step 2.** For each edge (u, v, t_1, t_2) in $\mathcal{F}_{t_s,t_e-1}$, if u is not connected to v via a path in $(V, \mathcal{F}_{t_s,t_e})$, we add it to $\mathcal{F}_{t_s,t_e}$.
  - **Step 3.** For each edge (u, v, t_s, t_e) in $\overline{\mathcal{G}}$, if u is not connected to v via a path in $(V, \mathcal{F}_{t_s,t_e})$, we add it to $\mathcal{F}_{t_s,t_e}$.



(a) Illustrating BuildAF **(b) Illustrating Lemma 3.6**

Figure 4: Illustrating the affiliated forests where each block corresponds to one affiliated forest. In Figure 4(a), the gray blocks are built in Phase-I while others are built in Phase-II following the ordering indicated by the arrows. In Figure 4(b), $\bar{e}$ is a spanning edge in $\mathcal{F}_{t_1,t_2}$ but not in $\mathcal{F}_{t_1,\tau(\bar{e})+1}$, and $\bar{e}$ appears in those gray blocks.

We then prove the correctness of the above construction by induction. For the base cases $\mathcal{F}_{t,t}$ with $0 \le t \le t_{\max}$, $(V, \mathcal{F}_{t,t})$ is clearly an affiliated forest based the construction in Phase-I. We assume that $\mathcal{F}_{i,j}$ with $j \le t_e - 1$ or $t_s + 1 \le i < t_e = j$ obtained in Phase-II (i.e., those preceding $\mathcal{F}_{t_s,t_e}$ in the ordering), including $\mathcal{F}_{t_s,t_e-1}$ and $\mathcal{F}_{t_s+1,t_e}$, are affiliated forests.

For the induction step, we show that $(V, \mathcal{F}_{t_s,t_e})$ is an affiliated forest by contradiction. Suppose that $(V, \mathcal{F}_{t_s,t_e})$ is not an affiliated forest, and let (V, T) be a spanning forest of $\overline{G}_{[t_s,t_e]}$. We can easily derive that $\mathcal{F}_{t_s,t_e}$ does not have multiple spanning edges between any pair of two vertices and its projected graph over $[t_s, t_e]$ is a forest since (1) edges from $\mathcal{F}_{t_s+1,t_e}$ does not form any multiple edge or any cycle, and (2) an edge (u, v, t_1, t_2) or (u, v, t_s, t_e) in Steps 2-3 of Phase-II can be added to $\mathcal{F}_{t_s,t_e}$ only when u is not connected to v in $\mathcal{F}_{t_s,t_e}$. Hence, by assumption, we have $|\mathcal{F}_{t_s,t_e}| < |T|$ and thus there exists at least one pair of vertices u' and v' that are connected in T via an edge (u', v') but remain disconnected in $\mathcal{F}_{t_s,t_e}$. We derive the contradiction by showing that $\overline{G}_{[t_s,t_e]}$ does not have the edge (u', v') since (1) $\overline{G}_{[t_s+1,t_e]}$ (resp. $\overline{G}_{[t_s,t_e-1]}$) does not have the edge (u', v') since otherwise u' is connected to v' in $\mathcal{F}_{t_s+1,t_e}$ (resp. $\mathcal{F}_{t_s,t_e-1}$) and thus $\mathcal{F}_{t_s,t_e}$ based on the construction, and (2) there does not exist spanning edge (u', v', t_s, t_e) in $\overline{\mathcal{G}}$ since otherwise u' is connected to v' in $\mathcal{F}_{t_s,t_e}$.

We remark that (1) the set of affiliated forests is not unique and can be an arbitrary one, and (2) for those spanning edges that do not appear in the affiliated forests, they are redundant and can be pruned from the affiliated graph, thereby yielding a refined affiliated graph. The refined version of the affiliated graph shown in Figure 3 is presented in Appendix of the technical report [1], which involves fewer spanning edges and may lead to a more effective and efficient index structure. We then show how to build indices over those spanning edges in the affiliated forests.

3.2.2 AF-3DT *index structure and querying process.* Consider the affiliated forests built by BuildAF and a spanning edge $\bar{e} = (u, v, t_1, t_2)$ in the forests. We observe that $\bar{e}$ appears only in the following range of affiliated graphs, as visualized in Figure 4(b).

Lemma 3.6. *Given a spanning edge $\overline{e} = (u, v, t_1, t_2)$. Let $\tau(\overline{e})$ be the timestamp such that $\overline{e} \in \mathcal{F}_{t_1,\tau(\overline{e})}$ and $\overline{e} \notin \mathcal{F}_{t_1,\tau(\overline{e})+1}$. An affiliated forest $(V, \mathcal{F}_{x,y})$ built by* BuildAF *contains $\overline{e}$ if and only if*

$$0 \le x \le t_1,\ t_2 \le y \le \tau(\overline{e}). \tag{4}$$

Proof. We prove this in two dimensions. For the x-dimension, based on Step 1 of Phase-II in BuildAF, we observe that an affiliated forest $(V, \mathcal{F}_{x_1,t})$ is a superset of the following affiliated forest $(V, \mathcal{F}_{x_2,t})$, i.e.,

$$\forall 0 \le x_1 \le x_2 \le t \le t_{\max}, \mathcal{F}_{x_1,t} \supseteq \mathcal{F}_{x_2,t}. \tag{5}$$

For the y-dimension, based on Step 2 of Phase-II in BuildAF, we observe that if an affiliated forest $(V, \mathcal{F}_{t,y_1})$ does not contain $\overline{e}$, none of the following affiliated forests $\mathcal{F}_{t,y_2}$ will contain it, i.e.,

$$\overline{e} \notin \mathcal{F}_{t,y_1} \implies \forall y_1 \le y_2 \le t_{\max},\ \overline{e} \notin \mathcal{F}_{t,y_2} \tag{6}$$

Besides, we note that (u, v, t_1, t_2) is first added to an affiliated forest $\mathcal{F}_{t_1,t_2}$ by BuildAF either in Phase-I or in Step 3 of Phase-II (note that it cannot appear in any affiliated forest $(V, \mathcal{F}_{x,y})$ with $x > t_1$ or $y < t_2$ since $\mathcal{F}_{x,y}$ is a subset of $\{(u', v', t'_1, t'_2) \in \overline{\mathcal{E}} \mid [t'_1, t'_2] \subseteq [x, y]\}$). Finally, based on Equation (5) and Equation (6), we can easily prove the correctness of the lemma. □

Based on the above lemma, each spanning edge $\overline{e} = (u, v, t_1, t_2)$ in the affiliated forests corresponds to a 3-D vector $\langle t_1, t_2, \tau(\overline{e})\rangle$. Clearly, an affiliate forest $(V, \mathcal{F}_{t_s,t_e})$ can be retrieved by finding all those edges satisfying Equation (4), i.e., $t_s \le t_1 \le t_2 \le t_e \le \tau(\overline{e})$. To achieve this efficiently, after obtaining all 3-D vectors, we organize them via a 3-D range tree, which takes $O(\overline{m} \log(\frac{\log \overline{m}}{\log\log \overline{m}})^2)$ space [9]. We note that each 3-D vector corresponds to a data point (x, y, z) in the 3-D range tree, and the 3-D orthogonal query with $(q_{x_1}, q_{x_2}, q_{y_1}, q_{y_2}, q_{z_1}, q_{z_2})$, which aims to find all data points satisfying $q_{x1} \le x \le q_{x2} \wedge q_{y1} \le y \le q_{y2} \wedge q_{z1} \le z \le q_{z2}$, can be answered in $O(k + \log^2 \overline{m})$ where k is the number of output data points [9]. Finally, to answer a Spanning-CC query with $[t_s, t_e]$, we first retrieve the affiliate forest $\mathcal{F}_{t_s,t_e}$ by conducting a 3-D orthogonal query with $(t_s, t_e, t_s, t_e, t_e, t_{\max})$ on the 3-D range tree, and then find all CCs with at least two vertices in its projected graph. We refer to the above process as AF-3DT-Query.

Lemma 3.7. AF-3DT-Query *answers a query in $O(n + \log^2 \overline{m})$.*

Proof. This can be easily verified based on the fact that an affiliated forest containing at most $n - 1$ edges. □

3.2.3 Construction of AF-3DT. Given all 3-D vectors, a 3-D range tree can be constructed in $O(\overline{m} \log \overline{m})$ [9]. One remaining question is how to compute $\tau(\overline{e})$ for each edge in the affiliated forests (while filtering out other redundant edges in $\overline{\mathcal{G}}$). We note that it can be done by constructing all affiliated forests via BuildAF in $O(\alpha(\overline{m}) + \alpha(n)t_{\max}^2)$ where $\alpha(\cdot)$ is the inverse Ackerman function (detailed analysis can be found in Appendix of the technical report [1]). However, this approach is time-consuming due to the need to construct $O(t_{\max}^2)$ affiliated forests. To address, we instead introduce an efficient *affiliated forest decomposition* method called AFDecomp. The idea is to utilize the overlaps between different affiliated forests to avoid building each one from scratch. We observe

Algorithm 1: AFDecomp

Input: An affiliated graph $\overline{\mathcal{G}} = (V, \overline{\mathcal{E}})$
Output: The set of 3-D vectors $\langle t_1, t_2, \tau(\overline{e})\rangle$

1 Obtain the edge ordering $\langle \overline{e}_1, \overline{e}_2, ..., \overline{e}_{\overline{m}}\rangle$ based on $\omega_1(\cdot)$;
2 Initialize $\mathcal{S}^{(0)} \leftarrow \emptyset$, $D \leftarrow \emptyset$, $i \leftarrow 1$;
3 **foreach** *edge $\overline{e}_i = (u, v, t_1, t_2)$ in the ordering* **do**
4 **if** *u and v are not connected in $(V, \mathcal{S}^{(i-1)})$* **then**
5 $\mathcal{S}^{(i)} \leftarrow \mathcal{S}^{(i-1)} \cup \{\overline{e}_i\}$;
6 **else**
7 $\overline{e}_{min} \leftarrow$ the edge with the smallest value of $\omega_2(\cdot)$ among those in the path connecting u and v;
8 **if** $\omega_2(\overline{e}_i) > \omega_2(\overline{e}_{min})$ **then**
9 $\mathcal{S}^{(i)} \leftarrow \mathcal{S}^{(i-1)} \cup \{\overline{e}_i\} \setminus \{\overline{e}_{min}\}$;
10 $\tau(\overline{e}_{min}) \leftarrow t_2 - 1$, $D \leftarrow D \cup \{\overline{e}_{min}\}$;
11 **else** $\mathcal{S}^{(i)} \leftarrow \mathcal{S}^{(i-1)}$;
12 **foreach** *edge $\overline{e}$ in $\mathcal{S}^{(\overline{m})}$* **do** $\tau(\overline{e}) \leftarrow t_{\max}$;
13 **return** $\{\langle t_1, t_2, \tau(\overline{e})\rangle \mid \overline{e} \in \mathcal{S}^{(\overline{m})} \cup D\}$;

that for each spanning edge $\overline{e}$ in the affiliated forests, we have

$$\overline{e} \in \mathcal{F}_{0,\tau(\overline{e})} \text{ and } \overline{e} \notin \mathcal{F}_{0,\tau(\overline{e})+1}, \tag{7}$$

according to Equation (5). Therefore, to compute all τ values, we only need to construct $t_{\max}$ affiliated forests, i.e., $(V, \mathcal{F}_{0,0})$, $(V, \mathcal{F}_{0,1})$, ..., $(V, \mathcal{F}_{0,t_{\max}})$. Besides, these $t_{\max}$ affiliated forests also have some overlapping structures and thus can be built in an efficient *bottom-up* manner. Below, we elaborate on the details.

We first define two types of weights for each spanning edge $\overline{e} = (u, v, t_1, t_2)$ in $\overline{\mathcal{G}}$ as below.

$$\omega_1(\overline{e}) = (t_2 + 1) \times t_{\max} - t_1,\ \omega_2(\overline{e}) = (t_1 + 1) \times t_{\max} - t_2. \tag{8}$$

Let $\langle \overline{e}_1, \overline{e}_2, ..., \overline{e}_{\overline{m}}\rangle$ be a sequence of edges in $\overline{\mathcal{E}}$ called ω_1-edge ordering, which is sorted in a non-decreasing order w.r.t. $\omega_1(\cdot)$, i.e.,

$$\omega_1(\overline{e}_i) \le \omega_1(\overline{e}_j),\ 1 \le i \le j \le \overline{m}. \tag{9}$$

We note that one spanning edge (u, v, t_1, t_2) comes before the other (u', v', t'_1, t'_2) in the ordering if (1) $t_2 < t'_2$ or (2) $t_2 = t'_2$ but $t_1 > t'_1$.

We summarize AFDecomp in Algorithm 1. AFDecomp iteratively considers adding a spanning edge (u, v, t_1, t_2) to a graph $(V, \overline{\mathcal{S}})$, which is initially empty and globally maintained (as a spanning forest), based on a certain edge ordering $\langle \overline{e}_1, \overline{e}_2, ..., \overline{e}_{\overline{m}}\rangle$ (Lines 3-11). In particular, if u and v are not connected in $(V, \overline{\mathcal{S}})$, the spanning edge (u, v, t_1, t_2) is directly added to $\overline{\mathcal{S}}$ (Lines 4-5); otherwise, it is replaced with the spanning edge $\overline{e}_{min}$ with the smallest value of ω_2 among those edges in the path connecting u and v (Lines 6-10). We note that there exists some spanning edges that do not fall into the above two cases, i.e., u is connected to v in $(V, \overline{\mathcal{S}})$ and $\omega_2(\overline{e}) \le \omega_2(\overline{e}_{min})$ (Line 11). We remark that maintaining $\mathcal{S}$ only needs a single set and the notation $\mathcal{S}^{(i)}$ is used only to refer to the edge set $\mathcal{S}$ obtained after the i-th round at Line 3.

Correctness. We then show the correctness of AFDecomp. Recall that BuildAF could build different sets of affiliated forests. Based on the ω_1-edge ordering, we have the following definition.

Definition 3.8 (ω_1-based affiliated forests). The ω_1-based affiliated forests is a set of affiliated forests built by BuildAF based on the ω_1-edge ordering as described below.

- In Phase-I, $(V, E_{t,t})$ is the spanning forest of $\overline{G}_{[t,t]}$ with the smallest lexicographic order of $o_1 o_2 ... o_{|E_{t,t}|}$, where o_i is the index of edge $\overline{e}_{o_i}$ in the ω_1-edge ordering;
- In Phase-II, we consider each edge (u, v, t_1, t_2) or (u, v, t_s, t_e) to be added to $\mathcal{F}_{t_s,t_e}$ based on the ω_1-edge ordering.

We next show that AFDecomp returns the set of 3-D vectors for ω_1-based affiliated forests in the following lemma.

Lemma 3.9. AFDecomp *finds the set of 3-D vectors for the ω_1-based affiliated forests.*

Proof. Let $\{(V, \mathcal{F}_{0,0}), ..., (V, \mathcal{F}_{t_{\max},t_{\max}})\}$ be the set of ω_1-based affiliated forests. For any $0 \le y \le t_{\max}$, let $\overline{e}_{I(y)} = (u, v, t_1, t_2)$ with $0 \le I(y) \le \overline{m}$ be the last edge in the ω_1-edge ordering such that $t_1 = 0$ and $t_2 = y$. We observe that $\mathcal{F}_{0,y}$ $(0 \le y \le t_{\max})$ can be built by $\mathcal{S}^{(I(y))}$, formally,

$$\forall 0 \le y \le t_{\max},\ \mathcal{S}^{(I(y))} = \mathcal{F}_{0,y}. \tag{10}$$

The correctness of the above equation can be proved through induction, using a method similar to the one used to prove the correctness of BuildAF in Section 3.2.1. Based on Equation (10), those edges removed from $\mathcal{S}$ between the $I(y-1)$-th round and the $I(y)$-th round at Line 10, i.e., $\mathcal{S}^{(I(y-1))} \backslash \mathcal{S}^{(I(y))}$, are in $\mathcal{F}_{0,y-1}$ but not in $\mathcal{F}_{0,y}$. Then, by Equation (7), these edges have the τ value of $y-1$. For the remaining edges in $\mathcal{S}^{\overline{m}}$ (i.e., $\mathcal{F}_{0,\overline{m}}$), they have the τ value of $t_{\max}$. In summary, the correctness of AFDecomp can be guaranteed. □

Time complexity. We note that the iterative procedure corresponds to the problem of incremental maximum spanning forest maintenance, which can be solved using a link-cut tree structure implemented with common binary trees and splay trees [43]. Hence, the time complexity is bounded by $O(\overline{m} \log \overline{m})$.

Lemma 3.10. AFDecomp *runs in $O(\overline{m} \log \overline{m})$.*

3.3 An Advanced Index-based Solution: AF-BTAS

We note that 3-D orthogonal queries with $(t_s, t_e, t_s, t_e, t_s, t_{\max})$ conducted on AF-3DT involve only 2 input variables, i.e., t_s and t_e; and the 3-D range tree is designed for answering the general 3-D orthogonal queries with 6 input variables. Motivated by this, we propose to design a specialized index structure, called *Binary Tree with Auxiliary Substructures* (**BTAS**), for better answering the former queries (which finds all data points (x, y, z) such that $x \le t_s \le y \le t_e \le z$ for any input time window $[t_s, t_e]$). As a result, our affiliated-forest-based index solution with **BTAS**, named AF-BTAS (Affiliated-Forest-based BTAS Index), can be built in $O(\overline{m} \log \overline{m})$ and answers a spanning-CC query in $O(n + \log^2 \overline{m})$, which is the same as AF-3DT. We remark that AF-BTAS takes $O(\overline{m})$ space which is smaller than that of AF-3DT. Below, we elaborate on the details.

Index structure and construction. Given the set of 3-D vectors $P = \{(x_1, y_1, z_1), ..., (x_{|P|}, y_{|P|}, z_{|P|})\}$ (or, data points), our proposed **BTAS** T is a binary tree with auxiliary substructures. It can be built by recursively partitioning the set of data points into several smaller subsets, with each subset maintained in the corresponding subtrees or auxiliary substructures. Specifically, the **BTAS** T can be defined recursively as follows. See Figure 5 for illustration and Algorithm 2 for construction.

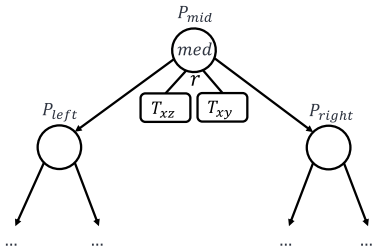


Figure 5: The recursive structure of BTAS

(1) If the data point set $P = \emptyset$, **BTAS** T is an empty tree;
(2) Otherwise, we define *med* by the median of the set of numbers $\{y_1, \ldots, y_{|P|}, z_1, \ldots, z_{|P|}\}$. We can partition the data points in P into three disjoint sets of data point P_{left}, P_{mid}, and P_{right}:

$$P_{left} := \{(x, y, z) \in P \mid z < med\},$$
$$P_{mid} := \{(x, y, z) \in P \mid y \le med \le z\},$$
$$P_{right} := \{(x, y, z) \in P \mid y > med\}.$$

Let the root node of the current **BTAS** T be r. Then, we can define:

- the left sub-tree T_{left} of r is a **BTAS** tree associated with the set of data points P_{left};
- the right sub-tree T_{right} of r is a **BTAS** tree associated with the set of data points P_{right}.

Moreover, the root node r stores the median *med* and is associated with the set of P_{mid} using two auxiliary substructures:

- T_{xy} facilitates querying 2-D data points in the set $\{(x, y) \mid (x, y, z) \in P_{mid}\}$, within the range $q_1 \le x \le q_2$ and $y \le q_2$.
- T_{xz} facilitates querying 2-D data points in the set $\{(x, z) | (x, y, z) \in P_{mid}\}$, within the range $q_1 \le x \le q_2$ and $q_2 \le z$.

We remark that the two auxiliary substructures in **BTAS** are implemented using the well-known *priority search tree*. In particular, a priority search tree takes input as three parameters q_x, q'_x, q_y and returns data points (x, y) that satisfy $q_x \le x \le q'_x$ and $q_y \le y$ in $O(\log a + b)$ time, where a is the total number of data points stored in the priority search tree and b is the number of reported data points [34]. We note that **BTAS** T is a balanced search tree, since it partitions P using the median *med*, resulting in two subtrees of comparable size. Moreover, the height of **BTAS** is $O(\log \overline{m})$. Finally, we analyze the time and space complexity as follows.

Lemma 3.11. *Algorithm 2 runs in $O(\overline{m} \log \overline{m})$ and* BTAS-Index *costs $O(\overline{m})$ space.*

Proof. This proof can be found in Appendix of [1]. □

Query processing. We summarize the query processing based on **BTAS** T in Algorithm 3. Given a time window $[t_s, t_e]$, based on the **BTAS** structure, we can efficiently identify the affiliated forest by finding data points (x, y, z) that satisfy the condition $t_s \le x \le y \le t_e \le z$ (Lines 1-2), and thus answer a spanning-CC query efficiently (Line 3). In particular, the procedure BTAS-Query-Rec corresponds to the procedure of finding data points from BTAS. In general, it descends recursively along a path in BTAS, which starts from the

Algorithm 2: `BTAS-Construction`

Input: The set P of n 3-D data points
Output: A **BTAS** tree T

$T \leftarrow \emptyset$;
if $P = \emptyset$ **then return** T;
$med \leftarrow$ the median of all y-coordinates and z-coordinates of data points in P, i.e., $\{y_1, \ldots, y_n, z_1, \ldots, z_n\}$;
$P_{left} \leftarrow \{(x, y, z) \in P \mid z < med\}$;
$P_{mid} \leftarrow \{(x, y, z) \in P \mid y \leq med \leq z\}$;
$P_{right} \leftarrow \{(x, y, z) \in P \mid y > med\}$;
left **BTAS** sub-tree of $T \leftarrow$ `BTAS-Construction`(P_{left});
right **BTAS** sub-tree of $T \leftarrow$ `BTAS-Construction`(P_{right});
$T_{xy} \leftarrow$ `PST-Construction`$(\{(x, y) \mid (x, y, z) \in P_{mid}\})$;
$T_{xz} \leftarrow$ `PST-Construction`$(\{(x, z) \mid (x, y, z) \in P_{mid}\})$;
return T;

Algorithm 3: `BTAS-Query`

Input: The **BTAS** structure T and a time window $[t_s, t_e]$
Output: All CCs in $\overline{G}_{[t_s,t_e]}$ with at least two vertices

1 $R \leftarrow$`BTAS-Query-Rec`(T, t_s, t_e);
2 Obtain the projected graph (V, E') of $(V, \mathcal{F}_{t_s,t_e})$;
3 **return** all CCs in (V, E') with at least two vertices;
4 **Procedure** **`BTAS-Query-Rec`**(T, t_s, t_e)
5 $\quad R \leftarrow \emptyset$;
6 $\quad$ **if** $T = \emptyset$ **then return** R;
7 $\quad med \leftarrow$ the median value stored in root node of T;
8 $\quad$ **if** $t_e \leq med$ **then**
9 $\quad\quad R \leftarrow R \cup$ `PST-Query`(T_{xy}, t_s, t_e);
10 $\quad\quad R \leftarrow R \cup$ `BTAS-Query`(T_{left}, t_s, t_e);
11 $\quad$ **else**
12 $\quad\quad R \leftarrow R \cup$ `PST-Query`(T_{xz}, t_s, t_e);
13 $\quad\quad R \leftarrow R \cup$ `BTAS-Query`(T_{right}, t_s, t_e);
14 $\quad$ **return** R;

root node to one of the leaf nodes. Along the path, it aggregates the sets of data points that meet the required condition in both the tree nodes and the auxiliary substructures. Upon reaching the leaf of T, the final set is returned. Specifically, starting with the root, we have the following cases.

(1) We first consider P_{left} and P_{right} in sub-trees of the current T.
   (a) If $t_e < med$, we search along the left sub-tree since any data point in P_{right} satisfies $y > med > t_e$ and thus violates the query condition $y \leq t_e$.
   (b) If $t_e > med$, we search along the right sub-tree since any data point in P_{left} satisfies $t_e > med > z$ and thus violates the query condition $t_e \leq z$.
   (c) If $t_e = med$, the data points in the final result set cannot lie in either sub-tree, and we handle these data points in the auxiliary substructures.
(2) We next consider P_{mid} and its two auxiliary substructures in the root node of the current T. In particular, we adopt the well-known priority search tree [34] to implement the auxiliary substructures; and `PST-Query` is the procedure for querying 2-D data points (details can be found in the technical report [1]).
   (a) If $t_e \leq med$, we know that $t_e \leq z$ is already satisfied for all data points in P_{mid}. Thus, we focus on the query condition $t_s \leq x \leq y \leq t_e$, and obtain the related data points using the auxiliary substructure T_{xy}.
   (b) If $t_e > med$, we know that $y \leq t_e$ is already satisfied for all data points in P_{mid}. Thus, we focus on the query condition $t_s \leq x \wedge t_e \leq z$, and obtain the related data points using the auxiliary substructure T_{xz}.

It is easy to verify the correctness of Algorithm 3 based on the above discussion. We then analyze the time complexity as below.

Lemma 3.12. `BTAS-Query` *answers a query in* $O(n + \log^2 \overline{m})$.

Proof. The time complexity is bounded by the part of finding data points from **BTAS** at Line 2. Specifically, Algorithm 3 would be recursively invoked $O(\log \overline{m})$ rounds and each round will visit a tree node since the height of a **BTAS** T is $O(\log \overline{m})$ and there are at most $\overline{m}$ data points in **BTAS**. Besides, for each round, the time cost is bounded by that of `PST-Query`, which runs in $O(\log \overline{m})$ time [34]. Finally, at most $n - 1$ data points will be returned since each corresponds to an edge in the affiliated forest $\mathcal{F}_{[t_s,t_e]}$. Therefore, the total complexity is $O(\log^2 \overline{m} + n)$. □

3.4 Summary and His-CSM Query Instances

Given a temporal graph $\mathcal{G}$ and a non-overlapping and monotone CSM $\mathcal{P}$, our framework involves three components, namely affiliated graph construction, index construction over affiliated forests, and query processing, as summarized in Table 1.

One remaining issue is *how to compute* $\overline{\mathcal{E}}$ for constructing $\overline{\mathcal{G}}$ based on $\mathcal{G}$ and $\mathcal{P}$. Let `OnlineQ`$\langle\mathcal{P}\rangle(G)$ be a procedure which receives a graph as the input and returns all CSMs in $\mathcal{P}(G)$. We note that its implementation has been widely studied for many CSMs, e.g., k-core [12, 23], (α,β)-core [29], and connected components [15]. Besides, we denote by $f(\mathcal{P}, G)$ the running time of `OnlineQ`$\langle\mathcal{P}\rangle(G)$. Based on this procedure, we present an algorithm `Construction` for constructing $\overline{\mathcal{G}}$ in Algorithm 4. The correctness can be easily verified. Besides, we can derive that (1) `Construction` runs in $O(|V|^2 \cdot t_{\max} \cdot f(\mathcal{P}, G))$ and (2) the number of spanning edges in $\overline{\mathcal{G}}$ is bounded by $|V|^2 t_{\max}$. We remark that `Construction` is a general method for various CSMs, which can be further improved for specific CSMs, as described in the following.

We then adopt our `AF-BTAS` for solving two His-CSM query instances, namely His-CC query and His-Core query, which have been studied in the literature [54, 60]. The corresponding index solutions are called **BTAS-CC** and **BTAS-Core**, respectively. We note that both CC and k-core are non-overlapping and monotone according to Lemma 2.3. Below, we provide the details.

`BTAS-CC` for His-CC query. Given a temporal graph $\mathcal{G} = (V, \mathcal{E})$, we observe that $\overline{\mathcal{G}} = (V, \{(u, v, t, t) \mid (u, v, t) \in \mathcal{E}\})$ is an affiliated graph according to Definition 3.3 since (1) each spanning edge (u, v, t, t) corresponds to a minimal u-v-TW, i.e., u and v are in the same CC in $G_{[t,t]}$, and (2) the constraint (that u and v are in the

Solution	CSM model	Affiliated graph construction	Index construction	Index space	Query processing
AF-3DT	CSM-free	$O(n^2 \cdot t_{\max} \cdot f(\mathcal{P}, G))$	$O(\overline{m} \log \overline{m})$	$O(\overline{m} \log(\frac{\log \overline{m}}{\log \log \overline{m}})^2)$	$O(n + \log^2 \overline{m})$
AF-BTAS	CSM-free	$O(n^2 \cdot t_{\max} \cdot f(\mathcal{P}, G))$	$O(\overline{m} \log \overline{m})$	$O(\overline{m})$	$O(n + \log^2 \overline{m})$
BTAS-CC	His-CC	$O(n + m)$	$O(\overline{m} \log \overline{m})$	$O(\overline{m})$	$O(n + \log^2 \overline{m})$
TSF [54]	His-CC	N.A.	$O(\overline{m} t_{\max})$	$O(\overline{m})$	$O(n + t_s)$
BTAS-Core	His-Core	$O(m^* \cdot \overline{t} \cdot D_{\max})$	$O(\overline{m} \log \overline{m})$	$O(\overline{m})$	$O(n + \log^2 \overline{m})$
PHC [60]	His-Core	N.A.	$O(m^* \cdot \overline{t} \cdot D_{\max})$	$O(m^* \cdot \overline{t})$	$O(n \cdot \log \overline{t})$

Table 1: Comparison among different algorithms, where m^* is the number of edges in $G_{[0,t_{\max}]}$, $\overline{t}$ is theoretically bounded by $t_{\max}$, and $D_{\max}$ is the maximum degree. Authors in [54] claim that TSF can answer the query in $O(n)$, which ignores the time costs of collecting edges in the index. Based on the implementation of TSF, it actually answers the query in $O(n + t_s)$.

Algorithm 4: Construction($\mathcal{P}, \mathcal{G}$)

Input: The temporal graph $\mathcal{G} = (V, \mathcal{E})$ and a CSM $\mathcal{P}$
Output: An affiliated graph $\overline{\mathcal{G}}$

```
1  Ē ← ∅;
2  foreach pair of vertices u and v in V do
3      [t_s, t_e] ← [0, 0];
4      while t_e is no larger than t_max do
5          P(G_[t_s,t_e]) ← OnlineQ⟨P⟩(G_[t_s,t_e]);
6          if u, v are in the same subgraph in P(G_[t_s,t_e]) then
7              while t_s ≤ t_e and u, v are in the same subgraph
                 in P(G_[t_s,t_e]) do [t_s, t_e] ← [t_s + 1, t_e] ;
8              Ē ← Ē ∪ {(u, v, t_s − 1, t_e)};
9          [t_s, t_e] ← [t_s, t_e + 1];
10 return Ḡ(V, Ē);
```

Datasets	$\lvert V \rvert$	$\lvert \mathcal{E} \rvert$	$t_{\max}$
contact (CT)	275	28,244	15,662
ColledgeMsg (CM)	1,862	59,835	58,910
sx-superuser (SU)	159,481	144,339	1,437,198
Emain-E (EE)	966	332,334	207,879
sx-mathoverflow (MO)	21,980	506,550	505,783
facebook-wosn-links (FL)	63,732	817,035	333,924
sx-askubuntu (AU)	127,047	964,437	960,865
mit (MI)	97	1,086,404	33,452
wiki-talk-temporal (WT)	1,120,716	7,833,140	7,375,041
youtube-u-growth (YG)	3,223,590	9,375,374	203
wikipedia-growth (WG)	1,870,710	39,953,145	2,198
sx-stackoverflow (SO)	2,296,666	63,497,050	52,423,236

Table 2: Statistics of all datasets

same CC in $G_{[t,t]}$ if and only if u is reachable to v in $\overline{G}_{[t_s,t_e]}$) is satisfied since $G_{[t_s,t_e]}$ is the same as $\overline{G}_{[t_s,t_e]}$.

We note that the affiliated graph $\overline{\mathcal{G}}$ can be efficiently obtained in $O(n+m)$, which is also sparser than that obtained by Construction. Then, we can construct the corresponding **BTAS** index structure in $O(\overline{m} \log \overline{m})$ by AFDecomp and BTAS-Construction (note that $\overline{m}$ is equal to m based on the built affiliated graph). Finally, we can answer the His-CC queries in $O(n + \log^2 \overline{m})$ by using BTAS-Query. We call the above adaptation BTAS-CC and summarize in Table 1. In particular, compared to the SOTA **TSF** [54] for answering His-CC query, our BTAS-CC has following benefits. First, BTAS-CC has the time cost for index construction and query processing practically and theoretically smaller than that of **TSF** (as verified in Figure 8). The possible reasons include (1) AFDecomp will filter out some redundant edges from the affiliated graph, and (2) **TSF** needs to iterate each timestamp from 1 to t_s for answering a His-CC query. Second, BTAS-CC has a space cost comparable to that of **TSF** in both theory and practice (as verified in the experiments).

BTAS-Core for His-Core query. The affiliated graph can be built by adopting existing historical k-core enumeration methods [44, 55]. Specifically, they can list all distinct k-cores along with their minimal time intervals in $O(m^* \overline{t} D_{\max})$, as summarized in Table 1. Then, we follow our index-based framework for building the index and answering the queries. We observe that our BTAS-Core outperforms the SOTA **PHC** theoretically and practically. For example, in the worst case, BTAS-Core has the space cost for indices bounded by $O(nt_{\max})$ and the time cost for query processing bounded by $O(n + \log^2(nt_{\max}))$, while **PHC** has those bounded by $O(n^2 t_{\max})$ and $O(n \log t_{\max})$ (note that $\overline{m} \leq nt_{\max}$ since there are at most $nt_{\max}$ edges in the affiliated forests).

4 EXPERIMENTS

We conduct extensive experiments to evaluate the performance of our proposed AF-BTAS. In particular, we adapt our AF-BTAS to His-CC and His-Core queries (described in Section 3.4) and compare the resulting adaptations, namely **BTAS-CC** and **BTAS-Core**, with the SOTA **TSF** [54] (for His-CC) and **PHC** [60] (for His-Core), respectively. All algorithms are implemented in C++, compiled with g++ -O2, and run on a Linux machine with an Intel Xeon 2.60GHz CPU and 256GB RAM. Our source codes are available at https://anonymous.4open.science/r/Querying-Cohesive-Subgraphs-in-Temporal-Graphs-C38D.

Datasets. We evaluate the algorithms on real-world datasets collected from SNAP (https://snap.stanford.edu/data/) and KONECT projects (http://konect.cc/networks/). We note that the datasets used in [60] and [54] are covered in our experiments. The statistics of all datasets are shown in Table 2.

4.1 Experiments for Historical-CC Queries

We evaluate the performance of His-CC algorithms in terms of query processing and index construction.

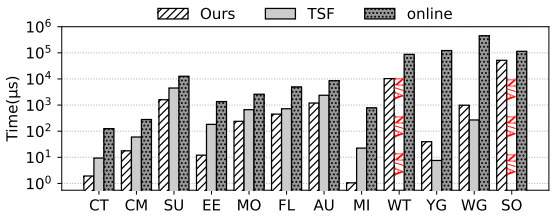


Figure 6: Average time cost per query for Historical-CC

Query processing. We generate 10,000 random queries with different time intervals of $[t_s, t_e]$ where both t_s and t_e are selected randomly. The average time cost per query on each dataset is shown in Figure 6. We have the following observations. **First**, our approach **BTAS-CC** is generally faster and achieves up to 20× speedups (on the EE and MI datasets) compared with the SOTA **TSF**. Moreover, we note that the results of the **TSF** index on the WT and SO datasets are marked as **N/A** since we are unable to complete the index construction within 24 hours. Besides, the **online** method does not need to construct the related index but shows the worst performance in terms of query processing. **Second**, the **TSF** index has a better performance on the YG and WG datasets. This may be due to the fact that the largest values of timestamp $t_{\max}$ in these two datasets — 203 for YG and 2,198 for WG — are relatively small compared with the scale of datasets (YG and WG have 9,375,374 and 39,953,145 temporal edges, respectively). In such cases, iterating through timestamps in the **TSF** approach becomes less time-consuming.

We also evaluate the time cost per query against different query time intervals, i.e., $t_e - t_s$, in Figure 7. Specifically, we set the interval sizes to as 20%, 40%, 60%, and 80% of $t_{\max}$ and record the average running time of 10,000 queries with each query interval size. We can see that our method **BTAS-CC** outperforms **TSF** and **online** in most cases, particularly when the size of query interval is small. For example, our method achieves up to 60× speedups on the EE and MI datasets when the interval size is 20% of $t_{\max}$. However, as the time interval size increases, the performance gap narrows, where **TSF** has a better performance on the query interval of $80\% t_{\max}$ in some datasets. This is because the query performance of **TSF** depends on the value of $t_{\max}$ and performs better when the query interval is large, say at least $80\% t_{\max}$. On the other hand, our approach **BTAS-CC** is suitable for different query time intervals.

Index construction. We first compare the efficiency of index construction on all datasets in Figure 8. As mentioned, the **TSF** method cannot construct the index within 24 hours on the WT and SO datasets. From Figure 8, it is clear to see that in most cases, our index construction method outperforms the **TSF** in terms of running time. In particular, we can construct our **BTAS-CC** index in 7.5 seconds on the WT dataset, while we are unable to build the **TSF** index even after spending more than 24 hours. The performance for index construction is in line with the theoretical result that the construction time of our method costs $O(\overline{m} \log \overline{m})$ while that of **TSF** requires $O(\overline{m} \cdot t_{\max})$ (as in Table 1). Besides, we report the space consumption of different methods in Figure 9. We can see that these two index structures **BTAS-CC** and **TSF** occupy nearly the same amount of space across most datasets, which is also implied by the same space complexity of $O(\overline{m})$ for both index structures.

4.2 Experiments for Historical-Core Queries

We evaluate the performance of His-Core algorithms in terms of query processing and index construction.

Query processing. We investigate the performance of query processing by varying the size of the input query time interval. Specifically, we first evaluate the average time cost for queries with 100,000 randomly selected pairs of t_s and t_e when $k = 30\% k_{\max}$ in Figure 10. From the results, we can see that (1) the **online** method cannot complete 100,000 queries in 12 hours for the WT, YG, and WG datasets; and (2) for the YG dataset, our method **BTAS-Core** runs 730× faster than the state-of-the-art **PHC** index.

Then, we vary the query interval size as 20%, 40%, 60%, and 80% of $t_{\max}$, and set the parameter k as 10%, 30%, 50%, 70% and 90% of $k_{\max}$. For each combination of query interval size and k, we randomly generate 100,000 queries. Figure 11 shows the average time cost per query for $k = 30\% k_{\max}$ (the results for other values of k are provided in Appendix of [1]). From Figure 11, we can observe that our method **BTAS-Core** outperforms the SOTA His-Core method **PHC** and the **online** method in all instances. For example, **BTAS-Core** outperforms **PHC** by at least 100 times across all parameter combinations tested on the YG dataset. Moreover, the query time of **BTAS-Core** increases as the interval size increases. This is because a larger interval size involves more vertices in the k-core, requiring **BTAS-Core** to search through more tree nodes in a **BTAS-Core** tree. We next conduct the experiments by varying the values of k on the FL and SU datasets in Appendix of [1]. We conclude that the query time of all indices decreases monotonically with increasing k, since a larger k implies fewer vertices in the resulting k-core.

Index construction. We report the time cost required for constructing the index structures for the datasets used in our experiments. Note that the construction methods for Historical-Core queries build the structures for all values of k at once. The results are shown in Figure 12. Both construction algorithms **BTAS-Core** and **PHC** involve computing changes in vertex core numbers over time windows, which is the most time-consuming step, leading to similar execution time for both methods. Note that the result on the SO dataset is not included as we are unable to construct both index structures within 24 hours. Additionally, we measure and report the space consumption of index structure in Figure 13. Both index structures have the same space complexity, but **BTAS-Core** generally occupies slightly more space than **PHC**.

5 RELATED WORK

Historical-CSM queries in temporal graphs. The problem of Historical-CSM queries in temporal graphs has attracted tremendous attention, which aims to design indices for efficient retrieval of CSMs within an arbitrary time interval. The k-core model has been a central topic in the study of Historical-CSM queries. Yu et al. [60] introduced the problem of historical k-core and proposed the PHC-Index to efficiently address the problem. Later, Yang et al. [56] developed the EF-index to support the query of historical k-core component, given a specific vertex and time interval. Additionally, Tian et al. [46] investigated the historical (α, β)-core model

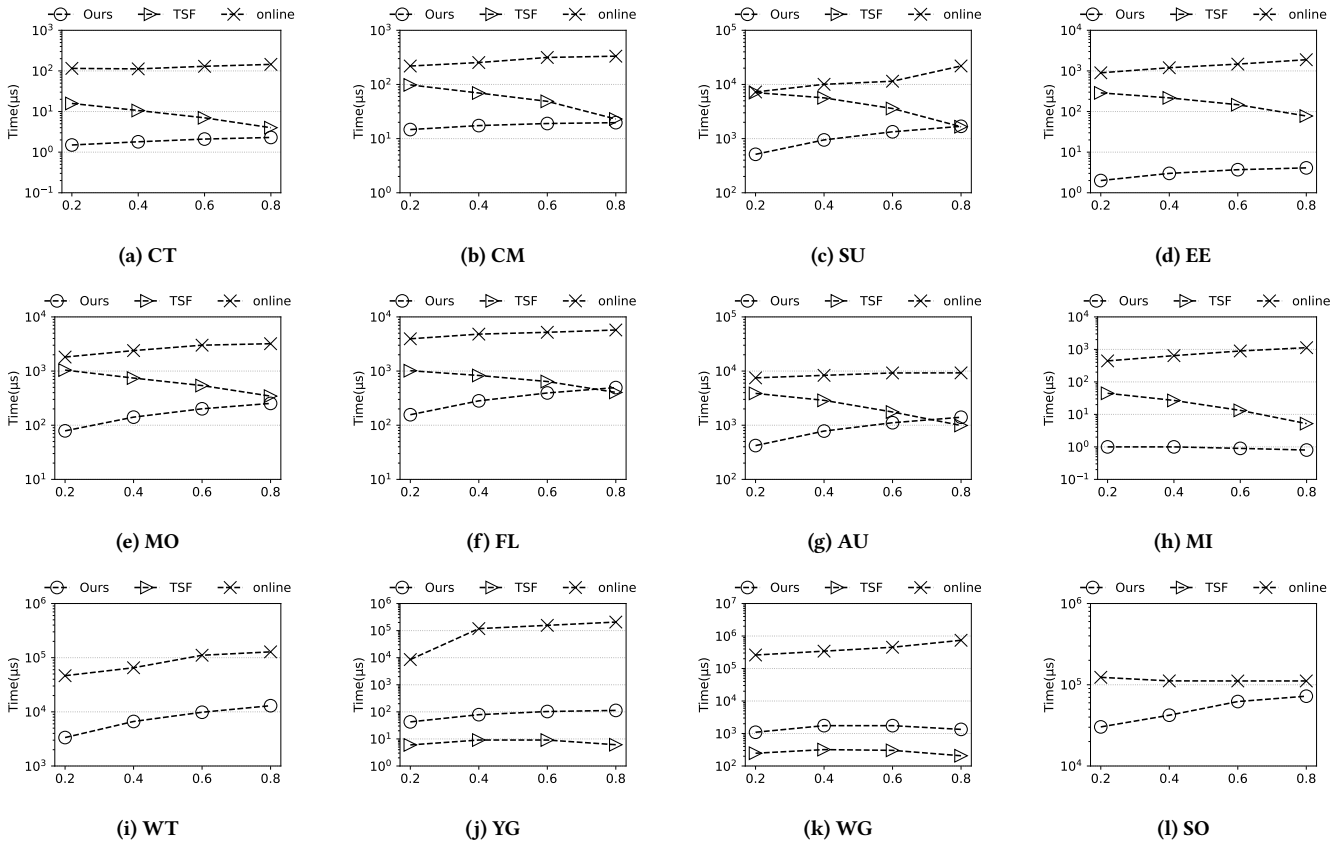


Figure 7: Historical-CC query time by varying query time interval sizes

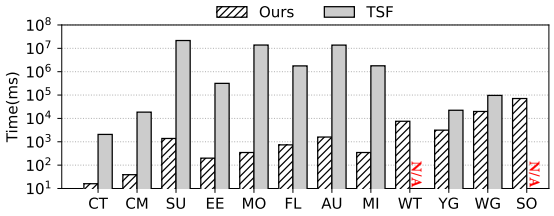


Figure 8: Index construction time for Historical-CC

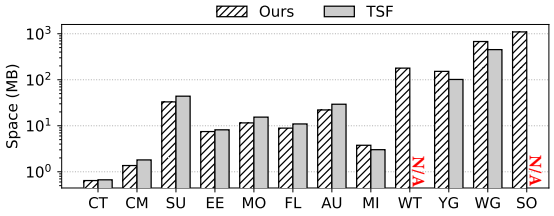


Figure 9: Index space consumption for Historical-CC

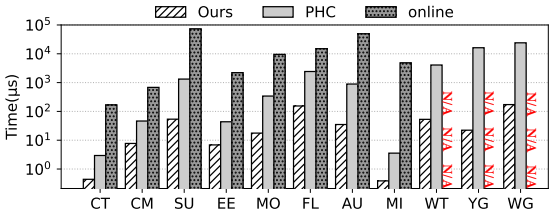
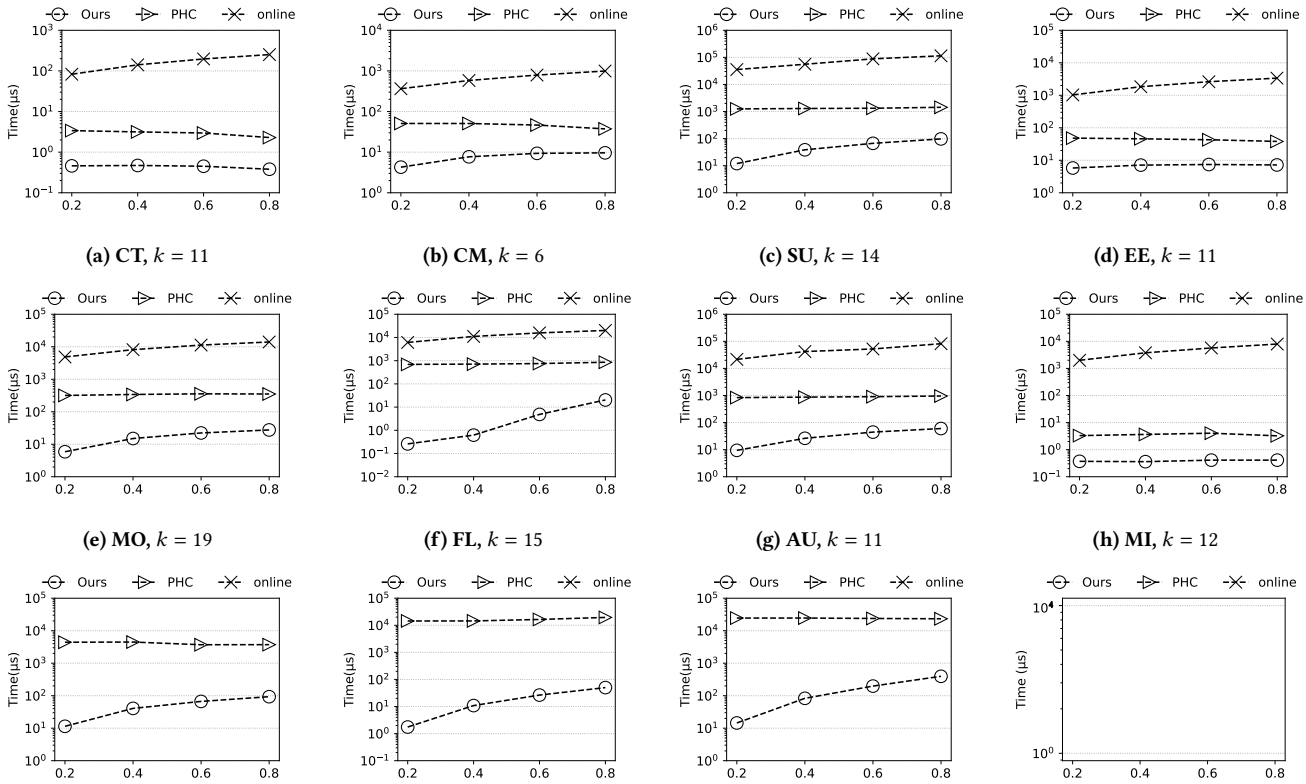


Figure 10: Average time cost per query with $k = 30\%k_{\max}$ for Historical-Core

in temporal bipartite graphs. Besides core-like structures, Xie et al. [54] focused on the (strongly) connected components and studied the corresponding historical queries in temporal graphs. Recently, Chen et al. [10] considered the historical queries for maximal clique.

Other CSM problems in temporal graphs. There also exist other CSM problems with different problem goals in temporal graphs. Yang et al. [55] and Zhong et al. [62] developed efficient online algorithms to identify all distinct historical k-cores, as well as k-cores that meet specific constraining conditions, across various time intervals. Galimberti et al. [18] introduced the concept of span-core, where two vertices must interact throughout the entire interval, and developed a core decomposition algorithm for this model. Similarly, Lotito and Montresor [30] introduced the concept of span-truss.



(a) CT, $k = 11$ **(b) CM,** $k = 6$ **(c) SU,** $k = 14$ **(d) EE,** $k = 11$

(e) MO, $k = 19$ **(f) FL,** $k = 15$ **(g) AU,** $k = 11$ **(h) MI,** $k = 12$

(i) WT, $k = 34$ **(j) YG,** $k = 26$ **(k) WG,** $k = 61$ **(l) SO,** $k = 45$

Figure 11: Historical-Core query time with $k = 30\% k_{\max}$ by varying query time interval sizes

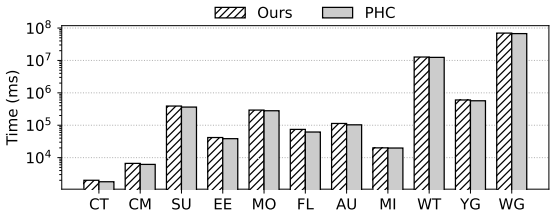


Figure 12: Index construction time for Historical-Core

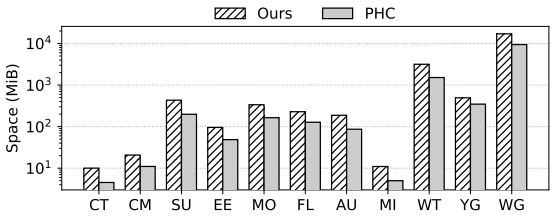


Figure 13: Index space consumption for Historical-Core

Other studies have explored various time-relevant metrics for k-cores in temporal graphs, such as interaction frequency [2, 52], persistence [25], burstiness [38], periodicity [39, 61], continuity [27], and reliability [45]. Besides, Guo and Sekerinski [19] studied how to maintain the core structure when a new edge is appended in the temporal setting, while Yang et al. [57] studied the quasi-clique in temporal graphs.

There are some other related studies include (1) the CSM-related problems on static graphs (e.g., k-plex [7, 8], k-core [3, 12], k-truss [14, 22, 47], γ-quasi-clique [59], k-defective clique [6, 16, 17], and densest subgraph [31]); and (2) queries (that are not related to CSMs) on temporal graphs (e.g., shortest path [51], historical connectivity between two vertices [44], reachability [42, 50], shortest path [51], communication motifs [20], and structural diversity [11]).

6 CONCLUSION

In this paper, we study the problem of historical non-overlapping and monotone cohesive subgraph models query (Historical-CSM), which includes the well-known Historical-CC and Historical-Core queries. To solve the problem, we propose two unified index-based framework, called `AF-3DT` and `AF-BTAS`. Finally, Extensive experiments are conducted and the results demonstrate the practical superiority of our method `AF-BTAS` over the state-of-the-art methods for answering Historical-CC and Historical-Core queries. In the future, we will explore the possibility of adapting our unified framework to other types of CSMs.

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A OMITTED PROOFS

A.1 Proof of Lemma 2.3

Proof. **k-core.** We first show that the k-core is a non-overlapping and monotone CSM. Recall that the k-core of a graph G is defined as a maximal subgraph where every vertex has degree at least k within the subgraph. To show that the k-core is non-overlapping, we will prove that the k-core is unique in a given graph G. Assume to the contrary that g_1 and g_2 are two different k-cores in a graph G. By the definition of k-core, the union of g_1 and g_2, i.e., $G[V(g_1) \cup V(g_2)]$, where $V(g)$ denotes the vertex set of subgraph g, is clearly k-core, which implies that there exists a larger k-core, contradicting the maximality of the k-core. Thus, k-core is a non-overlapping CSM. We next prove the monotone property. For any supergraph G' of G, we observe that for any vertex u, the degree of u in G' is at least the degree in G. Thus, any vertex that satisfies the k-core condition continues to satisfy the condition in G'. Thus k-core is a monotone CSM.

Connected component. We then show that the connected component is a non-overlapping and monotone CSM. Recall that a connected component of a graph G is a connected subgraph that is not part of any larger connected subgraph. Assume to the contrary that there are two connected components g_1 and g_2 that are overlapped, i.e., there exists a vertex $v \in V(g_1) \cap V(g_2)$. This implies that $v \in V(g_1)$ and $v \in V(g_2)$. Since $v \in V(g_1)$, by the definition of connected components, v is connected to every other vertex in $V(g_1)$ via a path. Similarly, since $v \in V(g_2)$, v is connected to every other vertex in $V(g_2)$ via a path. This implies that $G[V(g_1) \cup V(g_2)]$ forms a single connected component, contradicting the maximality of g_1 and g_2. Therefore, the connected component is non-overlapping. We next consider the monotone property. For any supergraph G' of G, a connected component C in G, C is still a connected subgraph in G', which indicates that C belongs to a larger connected component in G'. Thus, the connected component is a monotone CSM.

Besides, the non-overlapping and monotone of the connected k-core can be derived in a manner similar to the proof above. □

A.2 Time Complexity of `BuildAF` in Section 3.2.3

Proof. To perform the two phases of `BuildAF`, we make use of a classical disjoint-set (union-find) data structure [15]. The disjoint-set data structure with n elements is typically implemented using a forest of trees where each tree represents a set, and the root of each tree serves as the representative of its set, allowing for efficient operations to check connectivity in $\alpha(n)$ amortized time and merge sets in $O(1)$ time, where $\alpha(\cdot)$ is the inverse Ackerman function. Specifically, during Phase-I and Step 3 of Phrase-II, we perform connectivity checks for each spanning edges, resulting in an amortized time of $O(\alpha(\overline{m}))$ in total. Moreover, Step 1 of Phrase-II creates a copy of the disjoint-set structure, requiring $O(n)$ time. Then, Step 2 has at most n times of connectivity checks, incurring $O(\alpha(n))$ time. These two steps are executed for each $\mathcal{F}_{t_s,t_e}$, of which there are $O(t_{max}^2)$ in total. Consequently, the overall time complexity of the algorithm `BuildAF` is $O(\alpha(\overline{m}) + \alpha(nt_{\max}^2))$. □

A.3 Proof of Lemma 3.11

Proof. The **BTAS** structure is built recursively. For each recursion, the time complexity is dominated by the index construction time of the priority search tree T_{xy} and T_{xz} in Lines 9-10 of Algorithm 2. According to [34], the priority search trees T_{xy} and T_{xz} with $|P_{mid}|$ data points take $O(|P_{mid}| \log |P_{mid}|)$ time. Hence, the time in each recursion is $O(n + |P_{mid}| \log |P_{mid}|)$. Then, it is easy to see the total complexity of Algorithm 2 is $O(n \log n)$.

For the space complexity, we can observe that (1) each data point is only stored in one tree node in a **BTAS** tree, and (2) each data point is stored at most twice in the two priority search trees. Thus, the total space complexity is $O(n)$. □

B A RUNNING EXAMPLE OF REFINED AFFILIATED GRAPH

In Figure 14, we illustrate the refined affiliated graph derived from the affiliated graph shown in Figure 3. In this figure, each grid corresponds to a specific time interval, and each edge within a grid represents the spanning edge associated with that interval. The refined affiliated graph consists of a total of 12 spanning edges while the original affiliated graph has 19 spanning edges.

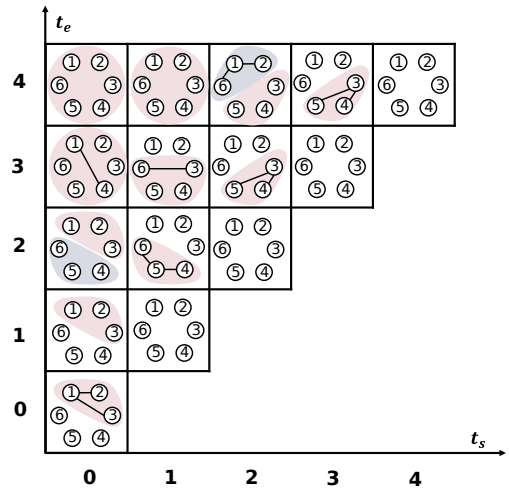


Figure 14: Illustrating the refined affilated graph

C DESCRIPTION OF PRIORITY SEARCH TREE

For completeness, we provide the related operations of priority search tree (PST) [34].

Given a 2-D point set P, the priority search tree can search points $(x, y) \in P$ that satisfy $x_{q1} \leq x \leq x_{q2} \wedge y_{q2} \leq y$. The standard procedures for the PST construction and query are shown in Algorithm 5 and Algorithm 6, respectively. We note that PST can also support three-sided queries such as $x_{q1} \leq x \leq x_{q2} \wedge y \leq y_{q2}$ in a similar manner. We also note that with a set of n 2-D data points, the corresponding PST can be constructed in $O(n \log n)$ time and answers each query in $O(k + \log n)$ time, where k is the number of returned data points [34].

Algorithm 5: `PST-Construction`

Input: The 2-D point set P
Output: A priority search tree T for P

$T \leftarrow \emptyset$;
$p \leftarrow$ the point in P with the minimum y-coordinates;
$P \leftarrow P \setminus \{p\}$;
$x_{med} \leftarrow$ the median of x-coordinates of points in P;
$P_{left} \leftarrow \{(x, y) \in P : x < x_{med}\}$;
$P_{right} \leftarrow \{(x, y) \in P : x > x_{med}\}$;
left sub-tree of $T \leftarrow$ `PST-Construction`(P_{left});
right sub-tree of $T \leftarrow$ `PST-Construction`(P_{right});

Algorithm 6: `PST-Query`

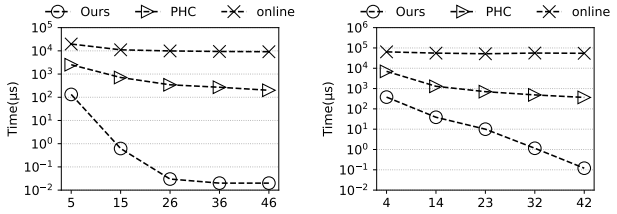
Input: The priority search tree T and the query parameters x_{q1}, x_{q2}, y_q
Output: The result set R of data points

$R \leftarrow \emptyset$;
if $T = \emptyset$ **then**
  return R;
$(x, y) \leftarrow$ the point with minimum y stored in the root node;
$x_{med} \leftarrow$ the median of x stored in the root node;
if $y > y_q$ **then**
  return R;
if $x_{q1} \leq x \leq x_{q2}$ **then**
  $R \leftarrow R \cup (x, y)$;
if $x_{q1} \leq x_{med}$ **then**
  $R \leftarrow R \cup$ `PST-Query`$(T_{left}, x_{q1}, x_{q2}, y_q)$;
if $x_{med} \leq x_{q_2}$ **then**
  $R \leftarrow R \cup$ `PST-Query`$(T_{right}, x_{q1}, x_{q2}, y_q)$;
return R;

D ADDITIONAL EXPERIMENTAL STUDIES FOR HIS-CORE

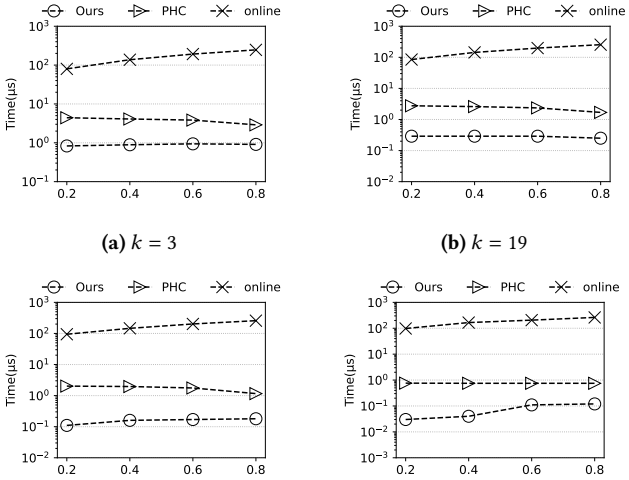
We first conduct the experiments by varying the values of k on the FL and SU datasets (shown in Figure 15). We can conclude that the query time of all indices decreases monotonically with increasing k, since a larger k implies fewer vertices in the resulting k-core.

We next provide additional experimental results for omitted values of k when k = 10%, 30%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core queries. Note that we focus on the experiments for query processing, since the index construction algorithms for Historical-Core queries build the index with all values of k. The results are illustrated in Figure 16 to Figure 26. We can conclude that our algorithm consistently outperforms the **PHC** and **online** methods across various parameter settings. Additionally, as k increases, the query performance of ours and **PHC** indexing structures improves, since the resulting k-core involves fewer vertices. For example, on the WG dataset (in Figure 26), **BTAS-Core** can solve queries within 1 microsecond (10^{-6} seconds) when k = 185, which is significantly faster (around 874,000× faster) than **PHC**.



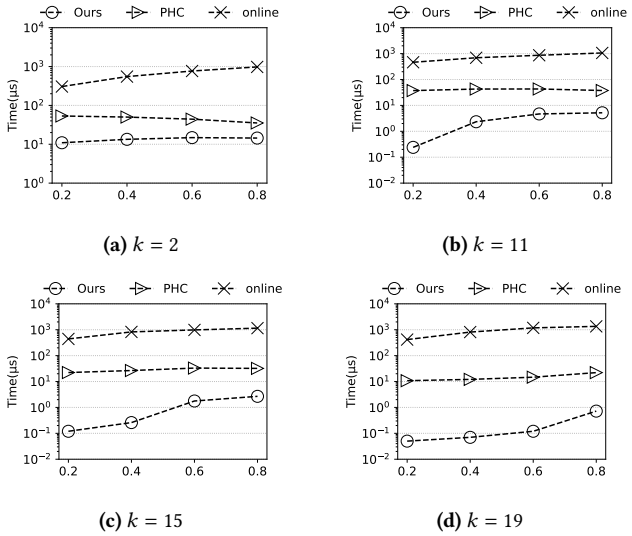
(a) FL **(b) SU**

Figure 15: Average time cost per query with the query interval of 30%$t_{\max}$ for Historical-Core



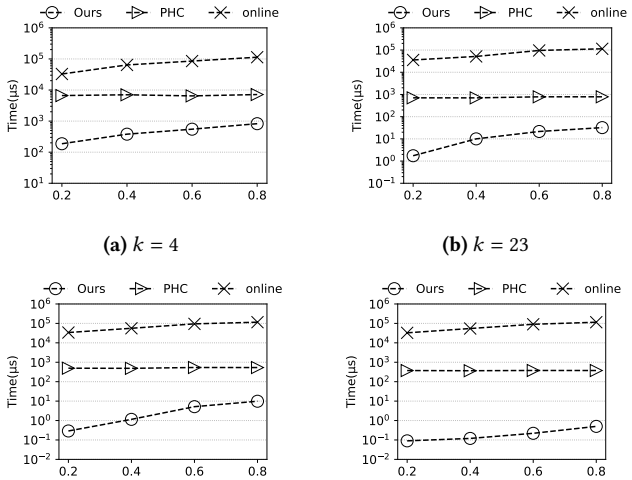
(a) $k = 3$ **(b)** $k = 19$ **(c)** $k = 27$ **(d)** $k = 35$

Figure 16: CT with k = 10%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core



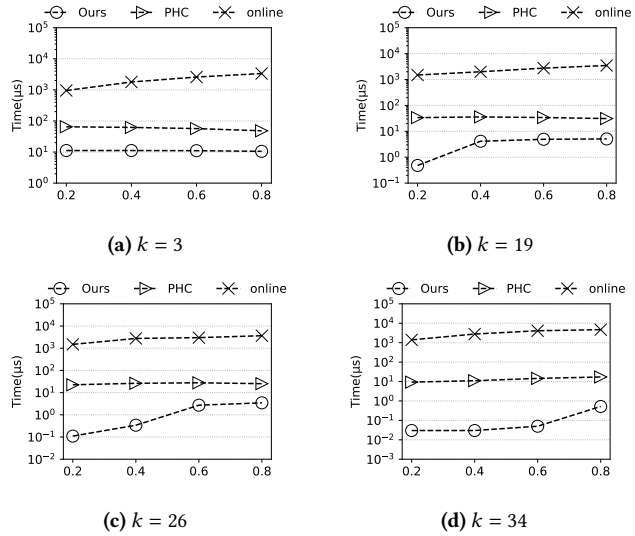
(a) $k = 2$ **(b)** $k = 11$ **(c)** $k = 15$ **(d)** $k = 19$

Figure 17: CM with k = 10%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core



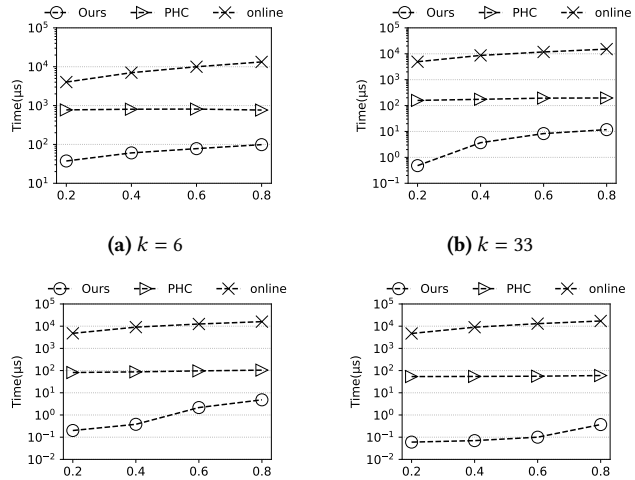
(a) $k = 4$ **(b)** $k = 23$ **(c)** $k = 32$ **(d)** $k = 42$

Figure 18: SU with k = 10%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core



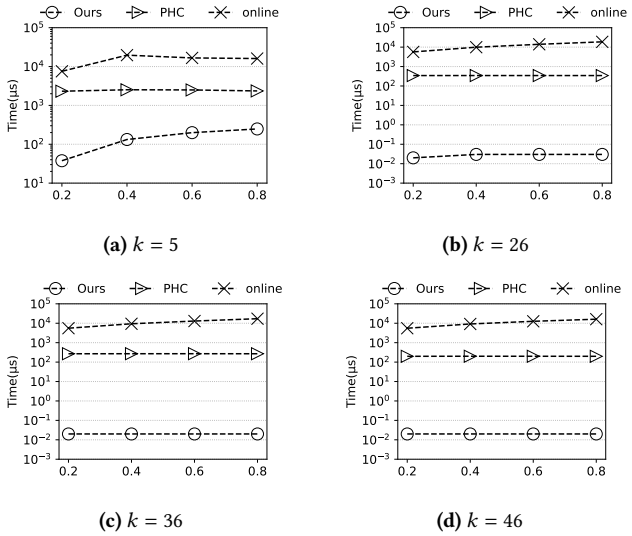
(a) $k = 3$ **(b)** $k = 19$ **(c)** $k = 26$ **(d)** $k = 34$

Figure 19: EE with k = 10%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core



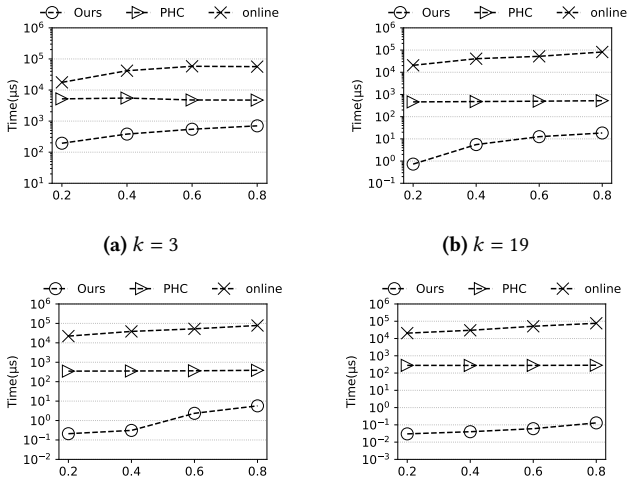
(a) $k = 6$ **(b)** $k = 33$ **(c)** $k = 46$ **(d)** $k = 59$

Figure 20: MO with k = 10%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core



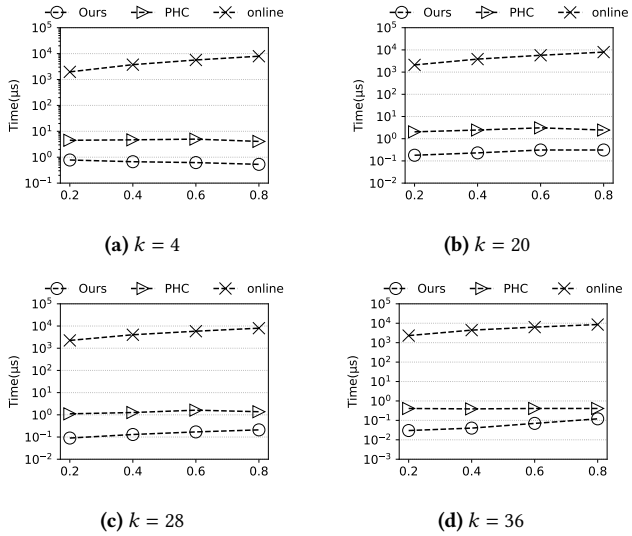
(a) $k = 5$ **(b)** $k = 26$ **(c)** $k = 36$ **(d)** $k = 46$

Figure 21: FL with k = 10%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core



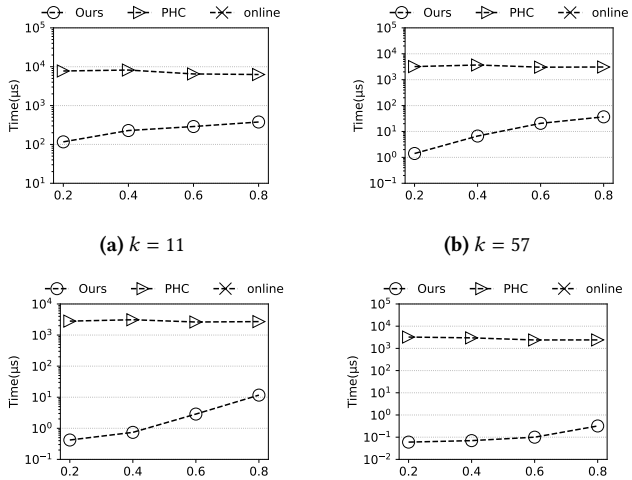
(a) $k = 3$ **(b)** $k = 19$ **(c)** $k = 26$ **(d)** $k = 34$

Figure 22: AU with k = 10%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core



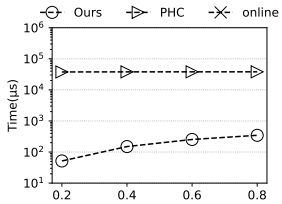
(a) $k = 4$ **(b)** $k = 20$ **(c)** $k = 28$ **(d)** $k = 36$

Figure 23: MI with k = 10%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core

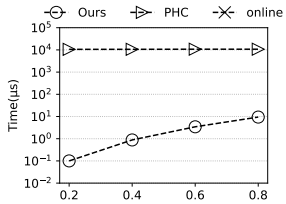


(a) $k = 11$ **(b)** $k = 57$ **(c)** $k = 80$ **(d)** $k = 103$

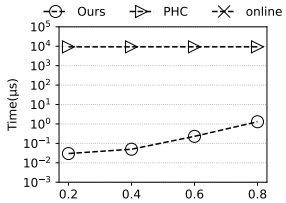
Figure 24: WT with k = 10%, 50%, 70%, and 90% of $k_{\max}$ for Historical-Core



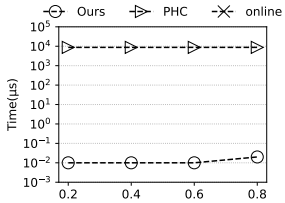
(a) $k = 8$



(b) $k = 44$

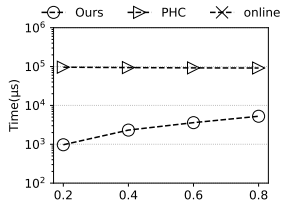


(c) $k = 61$

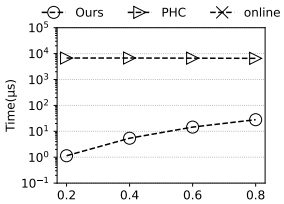


(d) $k = 79$

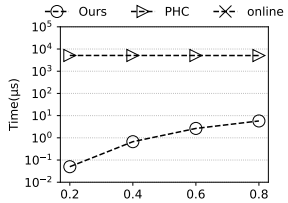
Figure 25: YG with $k = 10\%, 50\%, 70\%,$ and 90% of $k_{\max}$ for Historical-Core



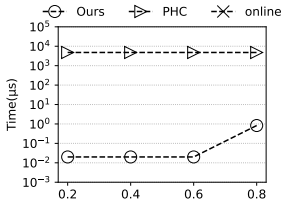
(a) $k = 20$



(b) $k = 103$



(c) $k = 144$



(d) $k = 185$

Figure 26: WG with $k = 10\%, 50\%, 70\%,$ and 90% of $k_{\max}$ for Historical-Core